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INTRODUCTION TO LOGIC

Contents

Preface

This is an introduction to logic based on a course taught at the University of Southern California. The book assumes no formal background or prior familiarity with logic. It is divided into three main parts. We first introduce the subject matter and main aims of logic and motivates the use of a formal language for that purpose. One block focuses on the language of propositional logic as a framework for the study of validity of an important class of natural language arguments. The language of quantificational logic is the subject of the final block, and its main purpose is to provide students with the toolkit they need in order to account for the validity of a wider family of natural language arguments.

The course aims to provide students with a number of formal methods designed in order to assess the validity of a variety of natural language arguments. These include the use of truth tables and a natural deduction system for propositional and quantificational logic. By the end of the course, students should be in a position to appreciate the relevance of formal methods for the study of language.

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Part I

Introduction

1

Reason and Argument

When we reason, we generally move from a body of statements, which we call *premises*, to another statement they support, which we call the *conclusion*. We will eventually refer to these inferences as *arguments*. We begin with the observation that some arguments appear to display a special feature: there is *no risk* that the conclusion may be false if the premises are true; given the form of the argument, the truth of the premises guarantees the truth of the conclusion. That seems to be the case with the first and fourth arguments below, but not with the others.

All Spanish citizens are EU citizens. No EU citizen requires a visa to enter Switzerland. Therefore, no Spanish citizen requires a visa to enter Switzerland.

Whales live underwater. Few mammals live underwater. Therefore, whales are not mammals.

Someone is carrying an umbrella on the street. So, it is raining.

No one carries an umbrella on the street unless it is raining. Someone is carrying an umbrella on the street. So, it is raining.

It is often difficult to tell whether the conclusion follows from the premises or the extent to which the argument is risk-free:

No human observer is able to observe more than a finite number of stars. So, humanity as a whole is only able to observe a finite number of stars.

If cobalt but no nickel is present in the sample, then a brown color should eventually appear. Either nickel or manganese is absent. Cobalt is present but only a green color appears. Therefore, nickel must be present.

The statue did not come into existence until we molded the clay of which it is made into one. However, the clay of which it is made existed well before we did that. If the statue is the same object as the clay, then whatever is true of one is true of the other. Therefore, the statue is not the same object as the clay; they are two distinct material objects occupying the same space at the same time.

Arguments

Arguments are composed of *declarative sentences*, which are sentences we use to make statements that are either true or false. We will be exclusively concerned with inferences from some declarative sentences, which are its premises, to a declarative sentence marked as its conclusion. More precisely:

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What is an argument?
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Definition 1.1. An *argument* is a sequence of declarative sentences. The last sentence in the sequence is its *conclusion*, and the sentences that precede the conclusion are its *premises*.

To avoid confusion, it is important to distinguish the technical use of the term in logic from more colloquial uses of the word.[1]. It will be helpful to consider some examples.

[1] Monty Python's Argument Clinic illustrates the difference.

Example 1.1. Consider the argument:

The laptop will not work if the battery is dead. The battery is dead. Therefore, the laptop will not work.

We represent the argument by means of a numbered sequence of statements. The last statement is the conclusion of the argument, and the statements that precede the conclusion are its premises.

1. The laptop will not work if the battery is dead.
2. The battery is dead.
3. The laptop will not work.

The conclusion of an argument is often preceded by transitional phrases such as 'therefore', 'thus', 'it follows that', etc. These phrases indicate that the statement in question is supported by other premises. Notice, however, that while it is perhaps usual for the conclusion of the argument to appear last in the argument, it is not uncommon for it to appear elsewhere in the argument.

Example 1.2. Consider the argument:

We do not know that the animal in the pen is a zebra. For if we did, then we would know it is not a cleverly painted mule. But we cannot be sure it is not a cleverly painted mule.

We represent the argument with the help of a list of premises followed by a conclusion:

1. If we knew that the animal in the pen is a zebra, then we would know it is not a cleverly painted mule
2. We cannot be sure it is not a cleverly painted mule.
3. We do not know that the animal in the pen is a zebra.

More generally, there is more than one way in which one and the same argument may be given in natural language:

The laptop will not work if the battery is dead. So, the laptop will not work. For the battery is dead.

The laptop will not work. After all, the battery is dead, and the laptop will not work if the battery is dead.

Validity and Form

We aim for a better characterization of validity, which we have tentatively characterized in terms of risk. On a first approximation, a valid argument is one where there is no risk that the premises may be true while the conclusion is false. That suggests a crude characterization of validity in terms of possibility:

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A false start
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An argument is valid if and only if it is *impossible* for the premises to be true while the conclusion is false.

That would be a false start, since the characterization would misclassify each of the two arguments below as valid:

The battery is dead. Therefore, the laptop will work or $\sqrt{2}$ is irrational.

The animal in the pen is a zebra. Therefore, there are infinitely many numbers.

While there is no risk for the premises to be true and the conclusion to be false in these cases, the reason has nothing to do with the form of the arguments but rather something to the subject matter of the statements involved. Instead, we will be concerned with cases in which it is the very form of the argument that explains the absence of risk.
Example 1.3. We may discern a common valid *argument form* in all three arguments given below:

1. If there is gas in the tank, then the engine will start.

2. There is gas in the tank.

3. Therefore, the engine will start.

1. If today is Thursday, then tomorrow will be Friday.

2. Today is Thursday.

3. Therefore, tomorrow will be Friday.

1. If the animal in the pen is a zebra, then it is not a cleverly painted mule.

2. The animal in the pen is a zebra.

3. Therefore, the animal in the pen is not a cleverly painted mule.

Here is the argument form in question:

1. If p, then q.

2. p

3. q

Other inferences exemplify different valid argument forms.

Example 1.4. All three arguments below exemplify a common valid *argument form*:

1. The battery is dead or the laptop is broken.
2. The battery is not dead.

3. Therefore, the laptop is broken.

1. Either the butler did it or the gardener did it.
2. The gardener did not do it.
3. So, the butler did it.

1. Either interest rates will be cut or inflation will continue to rise.
2. Interest rates will not be cut.
3. Therefore, interest rates will continue to rise.

Here is the form in question:

1. p or q

2. Not-p

3. q

We turn this observation into a crude definition of validity.

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What is a valid argument?
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Definition 1.2. An argument is *valid* if, and only if, it is an instance of a valid argument form.

One concern is that we have explained what is for an argument to be valid in terms of the exemplification of a valid *argument form.* But that just seems to postpone the question. We have made little progress unless we are able to explain what makes an argument form valid. That, however, is the one of the goals of the introduction of the formal languages we will subsequently study. They will give us the tools to identify and characterize a broad family of valid argument forms, and we will be able to explain their validity in terms of the behavior of logical connectives, which are formal counterparts of subsentential expressions in English. In the meantime, we will content ourselves with the fact that valid argument forms exemplify the following distinctive features:

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**Two distinctive features of valid argument forms**
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1. A valid argument form has *no* argument instances with true premises and a false conclusion.
2. A valid argument form preserves the truth of its premises regardless of the subject matter of its instances.

Let us look at further examples of valid arguments:
Example 1.5. Consider the argument:

If the mind the same thing as the brain, then the mind is mortal. The mind is immortal. Thefore, the mind is not the same thing as the brain.

In premise-conclusion form:

1. If the mind the same thing as the brain, then the mind is mortal.

2. The mind is immortal.

3. The mind is not the same thing as the brain.

This argument is an instance of a valid argument form:

1. If p, then q

2. Not q

3. Not p

A valid argument is not the same as a persuasive one. How persuasive a valid argument may be depends on how persuasive we find the premises of the argument. If the argument is valid, then the conclusion is guaranteed to be true if the premises are true. But a valid argument may contain false, and even utterly unpersuasive premises. That being said, we are especially interested in valid arguments with true premises, which is why we introduce a special label for those valid arguments.

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What is a sound argument?
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Definition 1.3. An argument is *sound* if, and only if, it is a valid argument and its premises are all true.

If an argument is sound, then the truth of the conclusion is guaranteed by the fact that the premises are all true and validity preserves truth.
Example 1.6. Compare the two arguments given below:

1. All cities in Southern California enjoy a warm weather.

2. Los Angeles is a city in Southern California.
3. Therefore, Los Angeles enjoys a warm weather.

1. All cities in Northern Europe enjoy a warm weather.
2. Barcelona is a city in Northern Europe.

3. Therefore, Barcelona enjoys a warm weather.

Both arguments are valid as they exemplify one and the same valid form, yet only the first is a candidate for being a sound argument, since it consists of true premises and a true conclusion. The second argument, in contrast, is valid but *unsound* since not all of its premises are true.

Formal Languages

The question of validity boils down to the question of which argument forms are valid. The validity of certain argument forms is in turn connected to the behavior of certain expressions in English, e.g., connectives such as ‘not’, ‘and’, ‘or’ or ’if ..., then .... So, the study of logic will require us to make inroads in the study of natural language.

We will introduce two formal languages in order to be able to isolate and study valid argument forms connected to the semantic behavior of different sets of expressions. These formal languages are in some respects less complicated than natural languages such as English and Spanish. You may even conceive of them as models of natural languages in which we can isolate and study the semantic behavior of selected natural language expressions.

We should distinguish three different aspects of a language, whether natural or formal: syntax, semantics, and pragmatics.

- Syntax is concerned with the formal features of the expressions of a language regardless of what they mean. In a spoken natural language like English, certain sounds are combined into syllables and words, which are in turn parts of larger syntactic items such as sentences. The grammar of the language specifies what counts as word and how words may be combined into well-formed sentences of the language.
- Semantics is concerned with the interpretation of the language. In order to be a competent user of a natural language, one must not only be able to recognize a word or a sentence of the language, one must know what they mean.
- Pragmatics studies the role of context in communication. In order to understand what is communicated by the use of a sentence, one must acknowledge know not only what the sentence literally means, but how that meaning interacts with the intentions of the speaker and the common ground of the context in which it is uttered.

Pragmatics will not be relevant for the study of the formal languages we will subsequently introduce, but it will play a role in our discussion of the semantic behavior of connectives such as 'or' and 'if …, then …' in natural language and the extent to which they are captured by the semantics of the formal language.

Exercises

1. Identify the premises and conclusion of each of the following arguments.

   a. You did not turn the ignition. The car starts when you turn the ignition and the tank is full. But the tank is full and the car did not start.

   b. If the mind is the same as the brain, the whatever is true of one is true of the other. The brain is a physically extended object but the mind is not. So, the mind is not the same as the brain.

   c. The deal will fall through unless Smith is part of the negotiation. So, the deal will not be completed. For Smith just phoned to let us know she cannot join the negotiation.

   d. You will be admitted into the club just in case you apply for club membership and you are not perceived to be eager to join. Applying for club membership is regarded as a sign of eagerness. That is, if you apply for club membership, you will be perceived to be eager to join. So, you will not be admitted into the club.

2. Which of the following arguments are valid?

   a. It today is Tuesday, then today is not Wednesday. Today is Tuesday. So, today is not Wednesday.

   b. Today is Tuesday or Thursday. So, today is Tuesday

   c. Today is Tuesday or Thursday. Today is not Thursday. So, today is Tuesday

   d. I'm late on Tuesdays. I'm late again today. So, today is Tuesday

3. Justify the invalidity of each of the argument patterns below by providing an instance with true premises and a false conclusion.

   a. If p, then q or r.
   Not q.
   Therefore, r.

   b. p unless q.
   q.
   p.

   c. p if, and only if, both q and r.
   r.
   q.

4. True or false? Justify your answers.

   a. There are valid arguments with false premises and a true conclusion.

   b. There are invalid arguments with true premises and a true conclusion.

   c. There are valid arguments with a false conclusion.

   d. There are sound arguments with a false conclusion.

   e. If an argument is valid, then at least one of its premises is true.

Part II

Propositional Logic

2

Formal Language

The language of propositional logic will allow us to model and study the validity of a broad family of argument forms underwritten by the behavior of propositional connectives such as 'not', 'and' and 'or'. To describe the language of propositional logic, we will specify:

- a *syntax* for the formal language, which will include a vocabulary of symbols and a set of grammatical rules that specify which strings of symbols count as sentences of the language, and
- a *semantics* for the formal language, which will explain how to interpret the language and what is for a sentence to be true under such an interpretation.

But first, we need some preliminaries.

Quotation and Metavariables

In order to describe a language, whether formal or natural, we will have to be able to refer to its expressions. When the target is a natural language such as English, we use quotation marks in order to *mention* a linguistic expression rather than *use* it.

Example 2.1. The expression 'Los Angeles' is *used* in one of the two sentences below, and it is *mentioned* in the other:

Los Angeles is composed of more than eighty cities.

'Los Angeles' is composed of two words.

The contrast we seek is between the *use* of a natural language expression to mean what it generally means and the mere *mention* of that expression to mean the expression itself, not what it means in natural language. In the former example, the name in question is used in the first sentence to mean what it ordinarily means in natural language, i.e., a city in southern California, but the name is merely mentioned in the second sentence to mean the very linguistic expression composed of two words, e.g., 'Los' and 'Angeles'.[1]

[1] Abbott and Costello's Who's on First illustrates the risk of conflating use and mention.

Example 2.2. The Arabic numeral '1' is used in one of the sentences below and it is mentioned in the other:

'1' is an Arabic numeral for the successor of 0.

1 is the remainder of 7 divided by 3.

Quotation marks give us the means to mention a natural language expression and to discuss its linguistic features. But we sometimes seek to make generalizations over linguistic expressions. For example, we may want to express the fact that whenever something is an English sentence, the concatenation of the expression 'it is not the case that' and that sentence form another sentence of English. One way to do this is to use Greek letters as metavariables to range over sentences:

Example 2.3. Greek letters may be used as metavariables over sentences of English in the two sentences below:

If φ is a sentence of English, then the concatenation of the expression 'it is not the case that' and φ is another sentence of English.

If we let φ be the sentence 'water is H2O' then the expression 'it is not the case that φ' is the sentence 'it is not the case that water is H2O'.[2]

[2] To ease notation, we may use 'it is not the case that φ' as an abbreviation for the concatenation of 'it is not the case that' and φ.

We will reserve the use of quotation marks for expressions of natural language in order to avoid confusion between use and mention. That means that we will avoid using quotation marks for expressions of the formal language *unless there is a risk of confusion.* On the other hand, we will make frequent use Greek letters as metavariables for expressions of the formal language being introduced.

Syntax

We specify the syntax of propositional logic in two stages. We will first specify the vocabulary of the language, and we will then explain how to combine the symbols of the language in order to produce sentences of the language.

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Vocabulary
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The vocabulary of propositional logic contains three types of symbols:

Propositional Variables These are the lowercase letters p, q, r, s, and t

with or without numerical subscripts:

$$p, q, r, s, t$$

Connectives These symbols formal counterparts of the natural language connectives ‘not’, ‘or’, ‘and’, and ‘if ..., then ...’:

$$\neg, \vee, \wedge, \to$$

Parentheses There are two parentheses:

$$), ($$

Nothing else is a symbol of the language.

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Grammar
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The role of grammar for a language is to explain how to combine symbols of the vocabulary into sentences of the language. For propositional logic, we use three main clauses for the characterization of *formula*. They explain how to generate more complex formulas from simpler formulas with the help of a connective.

Formula We define what is for an expression to be a formula of propositional logic.

1. All propositional variables are *formulas*.

2. If φ and ψ are *formulas*, then each of

$$\neg\varphi, (\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \to \psi)$$

is a *formula*.

3. Nothing else is a *formula*.

It will be helpful to look at concrete uses of the characterization in order to justify the fact that certain sequences of symbols are, or not, formulas of propositional logic.

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<img src="images/ex 2.5.png">

The construction tree for the formula $(p \vee (q \to \neg r))$.
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Example 2.4. The expression below is a formula:

$$(p \vee (q \to \neg r))$$

Here is a brief justification:

By rule 2, the expression $(p \vee (q \to \neg r))$ is a formula if both the expression p and the expression $(q \to \neg r)$ are formulas.
By rule 1, the expression p is a formula.
By rule 2, the expression $(q \to \neg r)$ is a formula if both the expression q and the expression $\neg r$ are formulas.
By rule 1, the expression q is a formula.
By rule 2, the expression $\neg r$ is a formula if r is a formula.
By rule 1, the expression r is a formula.

Therefore, we conclude that $(p \vee (q \to \neg r))$ is a formula.

The definition may similarly be used to justify the fact that a given string of symbols is *not* a sentence of propositional logic.

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<img src="images/ex-2.6.png">


There is no available construction tree.
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Example 2.5. The expression below is *not* a formula of propositional logic:

$$\neg(p \to A)$$

By rule 2, the expression $\neg(p \to A)$ is a formula *if* the expression $(p \to A)$ is a formula.
By rule 2, the expression $(p \to A)$ is a formula *if* both the expression p and the expression A are formulas.
By rule 1, the expression p is a formula.
By rule 3, the expression A is *not* a formula, since it is neither a propositional variable nor the outcome of one of the procedures described by rule 2.

Therefore, by rule 3, $\neg(p \to A)$ is *not* a formula.

We have explained how to construct complex formulas from simpler ones by means of very strict formation rules. Such formulas may sometimes become difficult to parse and tedious to write. For ease of expression, we will adopt a further notational conventions which will allow us to simplify notation. This is *not* meant as a revision of the official characterization of formula as it is not part of the official syntax of the language. The point of the rule instead is to allow us to use certain expressions as abbreviations for the official formulas.

Notational Convention We may remove the outer parentheses from a formula that is not part of another formula.

Example 2.6. We are able to use the expression

$$p \to q$$

is an abbreviation for the formula

$$(p \to q)$$

Example 2.7. We are similarly able to remove the outer parentheses of the formula:

$$((p \lor q) \lor r)$$

to obtain:

$$(p \lor q) \lor r.$$

as an abbreviation of the first formula.

Exercises

1. Add quotation marks to the following sentences in order to obtain a true sentence of English. If there is more than one solution, make sure to explain why.

   a. Los Angeles names a city in Southern California.

   b. Los is part of Los Angeles.

   c. Snow is white is a grammatical sentence of English.

   d. What is monosyllabic but whatever is polysyllabic.

2. Determine whether each of the following expressions is a sentence of propositional logic.

   a. $\neg\neg A$

   b. $(p \to (\neg q))$

   c. (p)

   d. $(q \to (\neg\neg p \lor r))$

3. Determine whether each of the following expressions is an *abbreviation* for a sentence of propositional logic.

   a. $p \to (q \land r)$

   b. $p \to q \to r$

   c. $p \lor (q \land r)$

   d. $p \to (q \land r \to q)$

3

Semantics

We will use assignments of truth values to propositional variables in order to interpret the language of propositional logic.

Definition 3.1. An *assignment* A for propositional logic maps every propositional variable into exactly one truth value (T or F).

We will declare a sentence of propositional logic to be true or false relative to an assignment of truth values to the propositional variables.

Truth under an Assignment We define what is for a sentence to be true under an assignment A:

A propositional variable p is true relative to A if, and only if, the assignment maps p into T.

A negation $\neg\varphi$ is true relative to A if and only if φ is not true relative to A.

A conjunction $(\varphi \wedge \psi)$ is true relative to A if and only if φ is true under A and ψ is true relative to A.

A disjunction $(\varphi \vee \psi)$ is true relative to A if and only if φ is true under A or ψ is true relative to A.

A conditional $(\varphi \rightarrow \psi)$ is true relative to A if and only if φ is not true under A or ψ is true relative to A.

Truth Tables

We use truth tables to represent how the truth value of complex sentences depends on the truth values of their simpler sentences. The following truth tables summarize the semantic clauses for negation, conjunction, disjunction, and the material conditional.

Negation Given a sentence φ, the truth value of the negation $\neg\varphi$ under an assignment is a function of the truth value of φ under the assignment.

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**Truth table for $\neg$**
```

φ	$\neg\varphi$
T	F
F	T

Conjunction Given two sentences φ and ψ, the truth value of the conjunction $(\varphi \wedge \psi)$ under an assignment is a function of the truth values of φ and ψ under the assignment.

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**Truth table for $\wedge$**
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φ	ψ	$(\varphi \wedge \psi)$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction Given two sentences φ and ψ, the truth value of the disjunction $(\varphi \vee \psi)$ under an assignment is a function of the truth values of φ and ψ under the assignment.

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**Truth table for $\vee$**
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φ	ψ	$(\varphi \vee \psi)$
T	T	T
T	F	T
F	T	T
F	F	F

Conditional Given two sentences φ and ψ, the truth value of the conditional $(\varphi \to \psi)$ under an assignment is a function of the truth values of φ and ψ under the assignment.

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**Truth table for $\to$**
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φ	ψ	$(\varphi \to \psi)$
T	T	T
T	F	F
F	T	T
F	F	T

We may use these rules to calculate the truth values of complex sentences of propositional logic relative to every assignment of truth values to its propositional variables.

Example 3.1. We may calculate the truth value of the sentence

$$\neg(p \wedge q)$$

relative to every assignment in two stages, which correspond to the way in which the sentence has been constructed from simpler constituents:

p	q	$(p \wedge q)$	$\neg(p \wedge q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

That means that the sentence is false when both p and q are mapped into the truth value T and true otherwise.

Example 3.2. We may calculate the truth value of the sentence

$$\neg(q \rightarrow p)$$

relative to every assignment in two stages:

p	q	$(q \rightarrow p)$	$\neg(q \rightarrow p)$
T	T	T	F
T	F	T	F
F	T	F	T
F	F	T	F

That means that the sentence is true if p is false and q is true relative to an assignment and false otherwise.

Example 3.3. We may now combine the two truth tables in order to produce a truth table for the complex sentence:

$$\neg(p \wedge q) \vee \neg(q \rightarrow p)$$

p	q	$\neg(p \wedge q)$	$\neg(q \rightarrow p)$	$(\neg(p \wedge q) \vee \neg(q \rightarrow p))$
T	T	F	F	F
T	F	T	F	T
F	T	T	T	T
F	F	T	F	T

That means that the formula is true under an assignment unless it assigns the truth value T to both p and q., which is the situation depicted by the first row of the truth table.

`How to construct a truth table`

We want to develop a systematic procedure for constructing a complete truth table for a target sentence. We proceed in three steps:

1. We devote a column for each propositional variable occurring in a sentence followed by a column for each subsentence for the target sentence.

2. The number of propositional variables occurring in a sentence determines the number of rows for the truth table. Given n propositional variables, there are exactly 2^n types of assignment to consider.[1] We start with the *innermost* column devoted to a propositional letter and we alternate occurrences of 'T' and 'F' until we populate the entire column. That is, 2^0 occurrences of 'T' followed by 2^0 occurrences of 'F'. For the next column on the left, we alternate 2^1, e.g., 2 occurrences of 'T' with 2^1, e.g., 2 occurrences of 'F'. More generally, for the nth column on the left of the innermost column devoted to a propositional letter, we alternate 2^n occurrences of 'T' with 2^n occurrences of 'F'.

3. We populate each subsequent column for a subsentence of the target sentence as a function of the truth values for the relevant subsentences of it.

[1] That means that a truth table for a sentence containing 3 propositional variables will require 8 rows; one for a sentence containing 4 propositional letters will require 16 rows; etc.

Tautologies and Contradictions

Tautology A formula φ is a *tautology* if, and only if, φ is true under all assignments of truth values to propositional variables.

Example 3.4. The formula

$$(p \to q) \vee (q \to p)$$

is a tautology.

p	q	$(p \to q)$	$(q \to p)$	$(p \to q) \vee (q \to p)$
T	T	T	T	T
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

Contradiction A formula φ is a *contradiction* if, and only if, φ is false under all assignments of truth values to propositional variables.

Example 3.5. The formula

$$\neg(p \to q) \wedge (q \vee \neg p)$$

is a contradiction.

p	q	$\neg(p \to q)$	$(q \vee \neg p)$	$\neg(p \to q) \wedge (q \vee \neg p)$
T	T	F	T	F
T	F	T	F	F
F	T	F	T	F
F	F	F	T	F

Equivalence and Consistency

Equivalence Two formulas φ and ψ are *equivalent* if, and only if, they are true under exactly the same assignments of truth values to propositional variables.

Example 3.6. The formulas $p \to q$ and $\neg p \vee q$ are equivalent because they are true under exactly the same assignments.

p	q	$p \to q$	$\neg p \vee q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Consistency A set of formulas is *consistent* if, and only if, there is at least one assignment of truth values to propositional variables under which all of its members are true. Otherwise, the set is *inconsistent*.

Example 3.7. The set $\{p \to q, q \to r, r \to \neg p\}$ consists of three formulas: $p \to q$, $q \to r$ and $r \to \neg p$. Now, the set is consistent because there is at least one assignment under which they are all true.

p	q	r	$p \to q$	$q \to r$	$r \to \neg p$
T	T	T	T	T	F
T	T	F	T	F	T
T	F	T	F	T	F
T	F	F	F	T	T
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

Validity

Validity An argument is *valid* if, and only if, there is no assignment of truth values to propositional variables under which all of its premises are true while its conclusion is false. Otherwise, the argument is *invalid*.

Example 3.8. Consider the argument given below:

$p \vee q$
$p \to \neg r$
$\neg r \vee p$

To assess its validity, we use a truth table to check whether there is some assignment on which the premises are true and the conclusion false:

p	q	r	$p \vee q$	$p \to \neg r$	$\neg r \vee p$
T	T	T	T	F	T
T	T	F	T	T	T
T	F	T	T	F	T
T	F	F	T	T	T
F	T	T	T	T	F
F	T	F	T	T	T
F	F	T	F	T	F
F	F	F	F	T	T

The fifth row in the truth table corresponds to an assignment on which the two premises are true and the conclusion is false. So, we conclude that the argument is *invalid*.

Example 3.9. Consider the argument given below:

$p \vee q$
$\neg(p \wedge r)$
$r \to q$

To assess its validity, we use a truth table to check whether there is some assignment on which the premises are true and the conclusion false:

p	q	r	$p \vee q$	$\neg(p \wedge r)$	$r \to q$
T	T	T	T	F	T
T	T	F	T	T	T
T	F	T	T	F	F
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	F	T	F
F	F	F	F	T	T

There is in this case no assignment on which the two premises are true and the conclusion is false. So, we conclude that the argument is *valid*.

```
The Search for Counterexample Method
```

The method of truth tables is perfectly general, since it enables one to survey all assignments of truth values to the sentence letters involved in the argument. One issue, however, is that it may require one to draw very complex and cumbersome truth tables in cases in which more than 3 propositional variables are involved in the argument. A truth table for an argument in which 5 propositional variables occur requires 2^5, i.e., 32, rows; 2^6, i.e., 64 rows, if the argument involves 6 propositional variables, etc.

In those cases, we may proceed differently: we search for a counterexample. That is, we set out to construct an assignment on which the conclusion is false and the premises are true. There are two eventual outcomes to this procedure. If we succeed, then the argument is invalid; if the construction of the assignment breaks down, then the argument is valid.

Example 3.10. Consider the argument given below:

$p \wedge q$
$r \to \neg q$
$p \to r$

We proceed to outline an assignment that makes the conclusion false:

p	q	r	$p \wedge q$	$r \to \neg q$	$p \to r$
					F

That requires p and r to be true and false, respectively:

p	q	r	$p \wedge q$	$r \to \neg q$	$p \to q$
T		F			F

Once we do, we set out to make the premises true. In particular, we realize we must make q true in order to verify $p \wedge q$:

p	q	r	$p \wedge q$	$r \to \neg q$	$p \to r$
T	T	F	T		F

On that assignment, we find that $r \to \neg q$ comes out true as well.

p	q	r	$p \wedge q$	$r \to \neg q$	$p \to r$
T	T	F	T	T	F

So, we found an assignment on which all the premises are true and the conclusion is false. The argument is *invalid*.

Example 3.11. Consider the argument given below:

$p \to \neg(q \to r)$
$\neg(\neg p_1 \vee \neg p)$
$(r \to p_2) \wedge (p_2 \to q)$
$q \vee r$

We set out to make the conclusion false:

p	q	r	p_1	p_2	$p \to \neg(q \to r)$	$\neg(\neg p_1 \vee \neg p)$	$(r \to p_2) \wedge (p_2 \to q)$	$q \vee r$
								F

That requires q and r to be respectively false.

p	q	r	p_1	p_2	$p \to \neg(q \to r)$	$\neg(\neg p_1 \vee \neg p)$	$(r \to p_2) \wedge (p_2 \to q)$	$q \vee r$
	F	F						F

That in turn makes $(q \to r)$ true and $\neg(q \to r)$ false. In order to make the first premise true, we must make p false:

p	q	r	p_1	p_2	$p \to \neg(q \to r)$	$\neg(\neg p_1 \vee \neg p)$	$(r \to p_2) \wedge (p_2 \to q)$	$q \vee r$
F	F	F			T			F

But notice that the second premise will now come out *false* under the assignment. Since p is false, $\neg p$ will be true as well as the disjunction $\neg p_1 \vee \neg p$, which makes its negation false.

p	q	r	p_1	p_2	$p \to \neg(q \to r)$	$\neg(\neg p_1 \vee \neg p)$	$(r \to p_2) \wedge (p_2 \to q)$	$q \vee r$
F	F	F			T	F !		F

We conclude that there is *no* assignment under which the premises are true and the conclusion is false. So, the argument is valid.

Exercises

1. Add quotation marks to the following sentences in order to obtain a true sentence of English. If there is more than one solution, make sure to explain why.

   a. Los Angeles names a city in Southern California.

   b. Los is part of Los Angeles.

   c. Snow is white is a grammatical sentence of English.

   d. What is monosyllabic but whatever is polysyllabic.

2. Determine whether each of the following expressions is a sentence of propositional logic.

   a. $\neg\neg A$

   b. $(p \to (\neg q))$

   c. (p)

   d. $(q \to (\neg\neg p \vee r))$

3. Determine whether each of the following expressions is an *abbreviation* for a sentence of propositional logic.

   a. $p \to (q \wedge r)$

b. $p \to q \to r$

c. $p \vee (q \wedge r)$

d. $p \to (q \wedge r \to q)$

4. Use the truth table method to determine whether each of the following sentences is a *tautology*, a *contradiction*, or neither a tautology or a contradiction.

   a. $((p \vee q) \to \neg(p \to q))$

   b. $p \to (q \to p)$

   c. $q \to (q \to p)$

5. Use the truth table method to determine whether the following pairs of sentences are logically equivalent.

   a. $p \to q$, $q \wedge \neg(p \vee q)$

   b. $p \to p$, $q \to q$

6. Use the truth table method to determine whether the following sets of sentences are consistent.

   a. $\{p, q, \neg(p \wedge q)\}$

   b. $\{p \vee q, q \to r, \neg(p \wedge r)\}$

7. Use the truth table method to determine whether the following arguments are valid.

   a. $p \to q$
   $q \to r$
   r

   b. $q \vee (p \wedge r)$
   $\neg q$
   r

8. Use the search-for-counterexample method to determine whether the following arguments are valid.

   a. $p \to (q \vee r)$
   $r \to (p \to q)$
   p
   q

   b. $(p \to q) \wedge (q \to p)$
   $\neg(p \wedge r)$
   $\neg(r \to s)$
   $s \to q$

4

Translation

The framework of propositional logic will enable us to isolate a family of valid argument forms underwritten by the behavior of certain truth functional connectives of English. To that end, we will have to be able to translate from English into the formal language we have just introduced.

A Translation Method

We will now develop the skill of translation from natural language to propositional logic. To that end, we break down the task into three main steps.

```
Step 1. Find a paraphrase
```

Paraphrase We paraphrase the relevant English sentence in order to make clear how the target sentence may be built from simpler constituents by means of connectives such as 'not', 'and', 'or', and 'if ..., then ...'

This first step will often require us to rewrite the sentence to include these connectives as opposed to some of their stylistic variants. We will use parentheses in order to highlight how the sentence has been constructed from simpler sentences.

Example 4.1. Consider the English sentence:

Although Smith is not a flexible negotiator, she will come to an agreement if we lower our demand.

We substitute 'and' for the stylistic variant 'although', and we rewrite the second part of the sentence to reveal how it is constructed from simpler sentences with the help of a conditional:

(It is not the case that Smith is a flexible negotiator, *and* (*if* we lower our demands, *then* she will come to an agreement)).

```
Step 2. Substitute the connectives:

 Substitute $\neg$ for 'not'

 Substitute $\wedge$ for 'and'

 Substitute $\vee$ for 'or'

 Substitute $\to$ for 'if $\dots$, then $\dots$.'
```

Substitute the connectives for their counterparts We replace the occurrences of the connectives we have identified above with the symbols for their counterparts in propositional logic.

Example 4.2. Consider the outcome of the first step above:

(It is not the case that Smith is a flexible negotiator, and (if we lower our demands, she will come to an agreement))

We replace each occurrence of a connective with an occurrence of its formal counterpart:

($\neg$ Smith is a flexible negotiator $\wedge$ (we lower our demands $\to$ she will come to an agreement)).

```
Step 3. Substitute a propositional variable for each simple sentence
```

Replace each simple sentence with a propositional variable We choose a propositional variable for each simple sentence and we proceed to substitute an occurrence of the propositional variable for each occurrence of the simple sentence in the target sentence.

Example 4.3. Consider the outcome of the second step above:

(It is not the case that Smith is a flexible negotiator, and (if Jones lowers her demands, she will come to an agreement))

```
*Translation key*\
$p$: Smith is a flexible negotiator.\
$q$: We lower our demands.\
$r$: Smith will come to an agreement.
```

We replace each simple sentence with a propositional variable in accordance with a suitable translation key.

$(\neg p \wedge (q \to r))$.

It will be helpful to look at a more complex example.

Example 4.4. Our target now is the English sentence below:

If cobalt is present, then either there is no nickel present or there is manganese present and a brown color appears.

We proceed in three steps:

Step 1 *If* Cobalt is present, *then* (it is *not* the case that there is nickel present *or* (there is manganese present *and* a brown color appears))

Step 2 (Cobalt is present → (¬ there is nickel ∨ (there is manganese ∧ a brown color appears)))

```
*Translation key*\
$p$: Cobalt is present.\
$q$: There is nickel present.\
$r$: There is manganese present.\
$s$: A brown color appears.
```

Step 3 $(p \to (\neg q \lor (r \land s)))$

Issues with Translation

Some of the examples above seem deceptively simple. We often find unexpected difficulties when we translate more idiomatic English sentences into the language of propositional logic. Let us distinguish a few different types of problems.

Non-Truth-Functional Connectives Some connectives resist paraphrase into a truth-functional connective.

Consider the English sentence:

My coffee had a sweet taste because I put sugar into it.

We may be tempted to paraphrase the sentence as a conjunction:

My coffee had a sweet taste *and* I put sugar into it.

But the paraphrase does not faithfully capture the truth conditions of the original sentence. More to the point, there is no reason to think the connective 'because' is truth-functional. In fact, we obtain a sentence with a different truth value when we replace one of its simpler parts, e.g., 'I put sugar into my coffee', with another sentence with the same truth value, e.g., 'Los Angeles is south of San Francisco':

My coffee had a sweet taste because Los Angeles is south of San Francisco.

To complicate matters even further, some uses of 'if ..., then ...' in the subjunctive are not truth-functional.

Consider the sentence:

If I had put motor oil in my coffee, it would have had a sweet taste.

Even if the sentence is false, notice that we may obtain a true sentence when we replace 'I had put motor oil in my coffee' with a sentence with the same truth value, e.g., 'I had put saccharine in my coffee'.

Structural Ambiguity A sentence may be structurally ambiguous, since there is more than one interpretation on which the sentence is built differently from different simpler sentences.

Consider the sentence:

Jones is at the office and Smith is on zoom or the deal will not be completed.

There are at least two different readings of this sentence depending on whether we regard it as a conjunction or as a disjunction:

Jones is at the office and (Smith is on zoom or the deal will not be completed)
(Jones is at the office and Smith is on zoom) or the deal will not be completed.

These two sentences come with different truth conditions, e.g., only the second sentence is true if Jones is *not* at the office but the deal is not completed.

Special Phrases Some English phrases present special challenges of their own.

The next issue arises from the question of how much force to apply to the original sentence in order to find a paraphrase constructed from more familiar truth-functional connectives with the same truth conditions as the original. We now look at several problem cases.

```
**$\dots$ only if $\dots$**
```

... *only if* ... I will win the lottery only if I buy a ticket.

Despite the superficial similarity between 'only if' and 'if', they achieve very different effects. Compare the sentence above with the conditional:

I will win the lottery *if* I buy a ticket.

This sentence tells us that the purchase of a ticket is a *sufficient* condition for a win and it may be paraphrased:

If I buy a ticket, then I will win the lottery.

The target sentence is quite different: it states that I must buy a ticket in order to have a chance to win the lottery. Otherwise put, the *only if* sentences says that the purchase is in fact a *necessary* condition for a win, and it should be paraphrased:

```
*Translation key*\
$p$: I will win the lottery.\
$q$: I buy a ticket.\
```

If I will win the lottery, then I will have bought a ticket.
$p \to q$

```
**$\dots$  if and only if $\dots$**
```

... *if and only if* ... The deal will be completed if, and only if, both parties agree to it.

It is not uncommon to combine 'if' and 'only if' into 'if and only if', which we should be able to parse as a conjunction:

```
*Translation key*\
$p$: The deal will be completed.\
$q$: Both parties agree to it.\
```

The deal will be completed if both parties agree to it, *and* the deal will be completed only if both parties agree to it.
$(q \to p) \land (p \to q)$

That means that both parties' agreement is both a *necessary* and *sufficient* condition for the deal to be completed.

We will sometimes use the symbol $\leftrightarrow$ in order to abbreviate formulas of that form. In general:

$$(\varphi \leftrightarrow \psi) := (\psi \to \varphi) \land (\varphi \to \psi).$$

It would be acceptable to use that symbol as an abbreviation as we translate from English into the language of propositional logic:

The deal will be completed if both parties agree to it, *and* the deal will be completed only if both partiess agree to it.
$(q \to p) \land (p \to q)$.
$(p \leftrightarrow q)$

```
**Neither $\dots$ nor $\dots$**
```

Neither ... nor ... Neither Jones nor Smith is in the office.

The goal now is to find a paraphrase constructed from familiar truth-functional connectives that is true if, and only if, the original sentence is true. There is more than one option at this point:

Jones is not in the office and Smith is not in the office.

It is not the case that Jones is in the office or Smith is in the office

In the first case, we turn the original sentence into a conjunction of two negations:

```
*Translation key*\
$p$: Jones is in the office.\
$q$: Smith is in the office.\
```

(It is not the case that Jones is in the office and it is not the case that Smith is in the office).
$(\neg p \wedge \neg q)$

In the second case, we paraphrase the original sentence into the negation of a disjunction:

It is not the case that (Jones is in the office or Smith is in the office).
$\neg(p \vee q)$

Both paraphrases are true in exactly the same circumstances, which coincide with the circumstances in which the target sentence is true. That is why both are adequate paraphrases of the original sentence.

```
**Unless**
```

Unless The deal will fall through unless Smith is in attendance.

The problem is to find a paraphrase constructed from familiar truth-functional connectives that is true in exactly the same circumstances as the original sentence. We have at least two options:

```
*Translation key*\
$p$: The deal will fall through.\
$q$: Smith is in attendance.\
```

If Smith is not in attendance, then the deal will fall through.
$\neg q \rightarrow p$

The deal will fall through or Smith is in attendance.
$p \vee q$

In the first case, we translate 'unless' as 'if not', whereas in the second case we translate it as 'or'. The paraphrases turn out to be equivalent in that they are true in exactly the same circumstances, which is what is wanted.

```
**Otherwise**
```

Otherwise Smith will be in attendance; otherwise, the deal will fall through.

The task now is to find a paraphrase constructed from familiar truth-functional connectives that is true in exactly the same circumstances as the original sentence. For example:

```
*Translation key*\
$p$: Smith is in attendance.\
$q$: The deal will fall through.\
```

Smith will be in attendance, and if Smith is not in attendance, then the deal will fall through.
$p \wedge (\neg p \rightarrow q)$

Application

The point of translation is to be able to discern valid argument forms in arguments formulated in English. In particular, we will often be interested in the question of whether a natural language argument exhibits a valid propositional form.
Example 4.5. Consider the natural language argument:

> I cannot be certain that the animal in the pen is a zebra unless I can rule out that it is not a cleverly painted mule. I cannot rule out that the animal in the pen is a cleverly painted mule. Therefore, I cannot be certain that the animal in the pen is a zebra.

To assess the argument, we follow three steps.

First, we put the argument into premise-conclusion form:

1. I cannot be certain that the animal in the pen is a zebra unless I can rule out that it is a cleverly painted mule.

2. I cannot rule out that the animal in the pen is a cleverly painted mule.

3. Therefore, I cannot be certain that the animal in the pen is a zebra.

Second, we translate the premises and conclusion of the argument into the language of propositional logic.

```
*Translation key*\
$p$: I can be certain that the animal in the pen is a zebra.\
$q$: I can rule out that the animal in the pen is a cleverly painted mule.
```

First premise:

I cannot be certain that the animal in the pen is a zebra unless I can rule out that it is not a cleverly painted mule.

If I can*not* rule out that the animal in the pen is a cleverly painted mule, *then* I can*not* be certain that the animal in the pen is a zebra.

($\neg$ I can rule out that the animal in the pen is a cleverly painted mule $\rightarrow$ $\neg$ I can be certain that the animal in the pen is a zebra).

$\neg q \rightarrow \neg p$

Second Premise:

I can*not* rule out that the animal in the pen is a cleverly painted mule.

$\neg$ I can rule out that the animal in the pen is a cleverly painted mule

$\neg q$

Conclusion:

I can*not* be certain that the animal in the pen is a zebra.

$\neg$ I can be certain that the animal in the pen is a zebra.

$\neg p$

We arrive at the translation of the argument into propositional logic:

1. $\neg q \rightarrow \neg p$
2. $\neg q$
3. $\neg p$

Third, we now use the search-for-counterexample method to determine whether the argument is valid in propositional logic:

We aim for an assignment that makes the conclusion false:

p	q	$\neg q \rightarrow \neg p$	$\neg q$	$\neg p$
T				F

In order to make the second premise true, we make q false:

p	q	$\neg q \rightarrow \neg p$	$\neg q$	$\neg p$
T	F		T	F

Unfortunately, the first premise comes out false under that assigment, which means that we cannot provide an assignment under which the premises come out true and the conclusion false.

p	q	$\neg q \rightarrow \neg p$	$\neg q$	$\neg p$
T	F	!	T	F

Let us move to an even more complex argument.

Example 4.6. Consider the natural language argument:

> You will be admitted into the club just in case you apply for club membership and you are not perceived to be eager to join. Applying for club membership is regarded as a sign of eagerness. That is, if you apply for club membership, you will be perceived to be eager to join. So, you will not be admitted into the club.

In order to assess the argument, we follow the same steps as before.

First, we put the argument into premise-conclusion form:

1. You will be admitted into the club just in case you apply for club membership and you are not perceived to be eager to join.
2. If you apply for club membership, you will be perceived to be eager to join.
3. You will not be admitted into the club.

Second, we translate the premises and conclusion of the argument into the language of propositional logic.

```
*Translation key*\
$p$: You will be admitted into the club.\
 $q$: You will apply for club membership.\
 $r$: You will be perceived to be eager to join.
```

First premise:

You will be admitted into the club if, and only if, (you apply for club membership and you are not perceived to be eager to join).

((you apply for club membership ∧ ¬ you are perceived to be eager to join) → you will be admitted to the club) ∧ (you will be admitted into the club → (you apply for club membership ∧ ¬ you are perceived to be eager to join)).

$(p \to (q \wedge \neg r)) \wedge ((q \wedge \neg r) \to p)$

Second Premise:

If you apply for club membership, you will be perceived to be eager to join.

You apply for club membership → you will be perceived to be eager to join.

$q \to r$

Conclusion:

You will not be admitted into the club.

¬ You will be admitted into the club.

$\neg p$

We arrive at the translation:

1. $(p \to (q \wedge \neg r)) \wedge ((q \wedge \neg r) \to p)$
2. $q \to r$
3. $\neg p$

Third, we use the search-for-counterexample method to determine whether the argument is valid in propositional logic:

We aim for an assignment that makes the conclusion false:

p	q	r	$(p \to (q \wedge \neg r)) \wedge ((q \wedge \neg r) \to p)$	$q \to r$	$\neg p$
T					F

In order to make the first premise true, we must make $q \wedge \neg r$ true, which requires q to be true and r to be false:

p	q	r	$(p \to (q \wedge \neg r)) \wedge ((q \wedge \neg r) \to p)$	$q \to r$	$\neg p$
T	T	F	T		F

Unfortunately, that makes the second premise false, which means that we cannot find an assignment under which the premises come out true and the conclusion false.

p	q	r	$p \to (q \wedge \neg r)) \wedge ((q \wedge \neg r) \to p)$	$q \to r$	p
T	T	F	T	!	F

Exercises

1. Translate the following sentences into the language of propositional logic.

   a. If interest rates fall, then we will be able to borrow more money and inflation will rise.

   b. Your computer will not start unless you turn it on and the battery is fully charged.

   c. The mayor's office made the decision, even though it did not sit well with its constituents.

   d. You will carry an umbrella outside just in case it is raining or you are under the impression that it is raining.

   e. The deal will be completed if all parties agree to it and the deal is found to be in compliance with current regulations.

2. Assess the following arguments for propositional validity.

a. The car will not start unless you turn the ignition and press the accelerator. You turned the ignition but the car failed to start. So, you did not press the accelerator.

b. The mind is a simple substance only if it is not made of further parts. If the mind is material, then it is is made of further parts. So, if the mind is a simple substance, then it is not material.

c. My client is not the burglar. For the doors show no signs of forced entry. If the doors show no signs of forced entry, then if my client is the burglar, he must have climbed to the unlocked window on the second floor. However, my client is arthritic and could not climb to that unlocked window on the second floor.

d. If the British are coming, then they are coming by land or by sea. Paul Revere will light only one lamp, if they are coming by land. He will light two lamps, if they are coming by sea. Paul Revere will light only one lamp. So, the British are coming.

e. The insurance company will cover your expenses only if you are current with your monthly payments. But if you are current with your monthly payments, then you can afford to cover the expenses yourself. So, the insurance company will not cover your expenses unless you can afford to cover them yourself.

5

Natural Deduction

We now develop a deductive system for propositional logic. The system will enable us to derive the conclusion of a valid argument from its premises in the language of propositional logic. These derivations will closely resemble chains of reasoning we may use to convince ourselves that a proposition follows from others in natural language. One difference, however, is that our derivations will display a very rigid structure where each step we take will be codified by a formal rule of inference. These rules of inference will in turn be closely tied to the semantic behavior of the connectives they formalize.

It will turn out that an argument formulated in the language of propositional logic is valid if, and only if, there is a derivation of the conclusion from its premises. That means that the deductive system will provide yet another approach to the study of validity in propositional logic.

There is a variety of different proof systems for propositional logic designed for different purposes. Some of them are specifically designed to facilitate the design of proofs in accordance with rules that are not dissimilar from steps we take in more familiar contexts, whereas others minimize the rules of inference available for a derivation in order to focus on the study of the metatheory of propositional logic. Here we will focus on a system of the first type otherwise known as a *natural deduction* system.

One way to approach an argument is to break it down into a series of elementary inferences that take us from the premises of the argument to its conclusion. These intermediate steps are licensed by rules of inference designed to either exploit or introduce complex formulas of a certain form. Such a chain of elementary inferences will be a *derivation*.

Derivation A *derivation* is a finite sequence of formulas, some of which may be marked as *premises* and *assumptions*. Other formulas in the sequence will be licensed by the application of a rule of inference to one or more prior formulas in the series. The last formula in the sequence will be its *conclusion*.

This preliminary characterization of a derivation will remain incomplete until we specify the rules of inference for the system, which we will do next. Once the rules of inference are in place, we will be able to check a purported derivation to make sure that each step accords to one of the rules and that the chain of inferences qualifies as a derivation.

Derivations will take the form of a numbered series of formulas that begin with the assumptions of the argument. We do this in order to be able to refer back to prior formulas in the proof, and we will draw a horizontal line after the premises. While formulas above the line are *assumptions*, the formulas that follow will be explicitly licensed by the rules of inference of the system or will on occasion be another assumption, which will be labelled as such.

Derivability A formula φ is *derivable from* a set of sentences Γ if, and only if, there is a *derivation* of φ whose non-discharged assumptions, if any, are all in Γ.

We will write $\Gamma \vdash \varphi$ to abbreviate the phrase 'φ is derivable from Γ.' If Γ is empty, we will just write $\vdash \varphi$. If Γ contains a finite number of sentences $\psi_1, \ldots, \psi_n$, we will write $\psi_1, \ldots, \psi_n \vdash \varphi$.

Natural Deduction Rules

Most natural deduction rules constrain either the introduction or the elimination of a connective in some complex formula of propositional logic. Each connective is governed by an introduction rule, which specifies the conditions under which a complex formula with that connective may be introduced into the derivation, and an elimination rule, which explains how to exploit a complex formula with that connective.

There is one important exception. We have a single structural rule, which enables us to repeat a formula from an earlier available line.

```
<img src="images/Repetition.png">\
  Repetition
```

Repetition You may repeat a formula φ on a line if it is available on a prior line of the derivation.

The annotation indicates that we are allowed to repeat the formula on line n at a later line m.

This is perhaps the simplest derivation you will ever encounter.

Example 5.1. $p \vdash p$

Conjunction and Conditional

We will now specify introduction and elimination rules for conjunction and the conditional, and we will postpone the discussion of similar rules for disjunction and negation until later.

```
<img src="images/conj intro.png">\
  Conjunction Introduction
```

Conjunction Introduction You may write a conjunction $\varphi \wedge \psi$ on a line if φ and ψ are available at prior lines in the derivation.

To illustrate the rule, notice how the introduction rule for conjunction suffices for a derivation of $p \wedge (r \wedge q)$ from three premises p, q, and r:

Example 5.2. $p, q, r \vdash p \wedge (r \wedge q)$

```
<img src="images/conj elim.png">\
  Conjunction Elimination
```

Conjunction Elimination You may write a conjunct φ on a line if a conjunction $\varphi \wedge \psi$ is available at a prior line

You may write a conjunct ψ on a line if a conjunction $\varphi \wedge \psi$ is available at a prior line

By way of illustration, note how the elimination rule for conjunction suffices for a derivation of r from three premises $p \wedge (r \wedge q)$:

Example 5.3. $p \wedge (r \wedge q) \vdash r$

```
<img src="images/cond elim.png">\
  Conditional Elimination
```

Conditional Elimination You may write ψ if the conditional $\varphi \rightarrow \psi$ and φ are available at prior lines.

We may now produce derivations for more sophisticated arguments.

Example 5.4. $p,\ p \rightarrow (q \wedge r),\ q \rightarrow (s \wedge t) \vdash r \wedge t$

```
<img src="images/cond intro.png">\
  Conditional Introduction
```

Conditional Introduction You may write $\varphi \rightarrow \psi$ if you are able to derive ψ from the assumption that φ. Once you do, you should bracket the lines employed in your auxiliary derivation of ψ from the assumption that φ and never ever appeal to them again.

The thought behind the rule is simple. We are entitled to write the conditional $\varphi \rightarrow \psi$ by itself in a line if we convince ourselves that we may eventually reach ψ on the assumption that φ.

Let us look at assorted applications of this rule:

Example 5.5. $p \to (q \wedge r) \vdash p \to (r \wedge q)$

Example 5.6. $p \wedge q \vdash p \to q$

Disjunction and Negation

The introduction and elimination rules for disjunction and negation require more care. We start with the rules for disjunction and postpone the discussion of negation until the end.

```
<img src="images/disj intro.png">\
  Disjunction Introduction
```

Disjunction Introduction You may write a disjunction $\varphi \vee \psi$ if a disjunct becomes available at a prior line.

```
<img src="images/disj elim.png">\
  Disjunction Elimination
```

The thought is that a disjunction must be the case if one of its disjuncts is the case. If Pluto is a planet, then Pluto is a planet or the Sun is a planet. For remember that it all takes for a disjunction to be the case is that one of its disjuncts is the case.

Disjunction Elimination You may write χ if the disjunction $\varphi \vee \psi$ and the conditionals $\varphi \to \chi$ and $\psi \to \chi$ are available at prior lines.

By way of motivation, notice that if $\varphi \vee \psi$ is the case, then there are two horns to consider. If $\varphi \to \chi$, then one horn leads to χ, and if $\psi \to \chi$, then the other horn leads to χ. But now, if both horns lead to χ, then we should be able to conclude χ on the basis of the disjunction $\varphi \vee \psi$ and the two conditionals above.

Here is an example of the elimination rule in action.

Example 5.7. $(p \wedge q) \vee (q \wedge p) \vdash p$

We finally introduce natural deduction rules for negation, which is the last connective left to consider. To reason with negation, it will be convenient to introduce a *new* symbol into the deductive system: $\perp$. This symbol, which is sometimes called *falsum* or *bottom* or *bot*, is designed to record the fact that we have arrived at a contradiction, and though it is not an official symbol of the language of propositional logic, it will play an auxiliary role in the formulation of the introduction and elimination rules for negation.

```
<img src="images/neg elim.png">\
  Negation Elimination
```

Negation Elimination You may write $\bot$ once a formula and its negation have become available in prior lines.

```
<img src="images/neg intro.png">\
  Negation Introduction
```

One way to understand the rule is as a permission to write the symbol for absurdity in the presence of a formula and its negation.

Negation Introduction You may write a negation $\neg\varphi$ if you are able to derive $\bot$ from the assumption that φ. Once you do, you should bracket the lines employed in your auxiliary derivation of $\bot$ from the assumption that φ and never ever appeal to them again.

Example 5.8. $p \vdash \neg\neg p$

This rule requires some explanation. What motivates it is the thought that we should conclude a negation of the form $\neg\varphi$ if we find that we are able to reach an absurdity on the assumption that φ is the case. How do we know that the square root of 2 is *not* rational? Because we are able to reach an absurdity from the assumption that $\sqrt{2}$ is a ratio of two integers $\frac{p}{q}$ without a common factor.[1]

[1] If $\sqrt{2} = \frac{p}{q}$, then $2 = \frac{p^2}{q^2}$ and $p^2 = 2q^2$. That means that p^2 is even, which requires p to be even. It follows that p^2 is divisible by 4 making q^2 and q even. So, p and q are both even, which means that they share a common factor. That is absurd, since we began with the assumption that they have no factors in common.

Example 5.9. $p \to q, \neg q \vdash \neg p$

```
<img src="images/EFSQ.png">\
  Ex Falso Sequitur Quodlibet
```

Ex Falso Sequitur Quodlibet You may write any formula once $\bot$ becomes available at a prior line.

The Latin phrase *Ex Falso Sequitur Quodlibet* means that anything at all follows from a contradiction, and we will use the initials EFSQ to abbreviate the rule that enables to write an arbitrary formula after a contradiction.

Example 5.10. $p \lor q, \neg p \vdash q$

Example 5.11. $\neg p \vdash p \to q$

We require one more rule in order to be able to cancel a double negation. That is, we want a rule designed to help us move from $\neg\neg\varphi$ to φ.

```
<img src="images/double negation.png">\
  Double Negation
```

Double Negation You may write a formula φ if a double negation $\neg\neg\varphi$ is available at a prior line.

Example 5.12. $\vdash p \vee \neg p$

That completes the list of natural deduction rules for propositional logic. They suffice for the purpose of deriving a conclusion from the premises of a valid argument of propositional logic.

How to Construct Proofs

There is no algorithm we can use in order to find a proof of a conclusion from a set of given premises, but sustained practice will help you develop a sense of how to proceed in each case. There are, however, some strategies to keep in mind when you first encounter a problem.

Your plan for a proof will require you to work backwards from what you want.

- If the conclusion of the argument is a conjunction $\varphi \wedge \psi$, then you should plan to be able to use the rule of conjunction introduction. In order to do that, you will aim to produce an auxiliary derivation of φ and an auxiliary derivation of ψ. Once they are in place, you will be able to write the conjunction by itself on a line citing the rule of conjunction introduction.

- If the conclusion of the argument is a conditional $\varphi \to \psi$, then you should generally plan to use the rule of conditional introduction. That is, you should plan to construct an auxiliary derivation of the consequent of the conditional from the assumption of the antecedent. The derivation in question will therefore begin with φ as an assumption and end with ψ as a conclusion. Given such a derivation, the rule of conditional introduction will license us to write the conditional on a line.

- If the conclusion is a negation of the form $\neg\varphi$, then you will plan to use the rule of negation introduction. In order to be able to apply it, you will start an auxiliary derivation of $\bot$ from φ as an assumption. Once the derivation is complete, you will be able to write the negation by itself on the basis of the rule of negation introduction.

Once you have devised a plan for a proof, you will work forward from your premises and/or assumptions in order to fill the gaps in the arguments. The elimination rules for each connective allow you to exploit them in order to complete some of the intermediate steps from the premises and assumptions to desired conclusions.

- Given a conjunction $\varphi \wedge \psi$, you may use the rule of conjunction elimination in order to write each conjunct by itself on a line, which may in turn be one of the intermediate steps required for the argument.

- Given a disjunction $\varphi \vee \psi$, you may use the rule of disjunction elimination, which is designed to capture a familiar type of reasoning. If we are able to prove a sentence χ from each disjunct, then we are entitled to write χ by itself as a line on the basis of the disjunction.
- To exploit a conditional $\varphi \to \psi$, we aim to find φ by itself on a line because we know that the rule of conditional elimination will enable us to write ψ by itself on a line.
- The case of negation $\neg\varphi$ is more delicate. We know that we are able to introduce $\bot$ by itself on a line if we are able to find an auxiliary derivation of φ, which becomes the new subgoal in order to exploit the original negation.

Exercises

1. Provide a natural deduction proof in order to justify each of the claims below:

   a. $p, \neg q \vdash p \to (\neg q \vee r)$

   b. $p \leftrightarrow (q \vee r), \neg q, \neg r \vdash \neg p$

   c. $p \to \neg(q \wedge r), q, \neg r \to \neg q \vdash \neg p$

   d. $p \leftrightarrow q, q \vee r, r \to (p \vee q) \vdash q$

   e. $(p \wedge q) \leftrightarrow (q \vee r), \neg r \to (\neg p \to q) \vdash p$

6

Limits of Propositional Logic

We sometimes speak of *the translation of an English sentence into propositional logic* when there is often more than candidate translation for the target sentence. There is, for example, room for the choice of different propositional variables and for the treatment of English phrases such as 'unless' or 'neither ... nor ...'. Part of the reason we ignore the differences between them is because they will be unimportant for our purposes. If an English sentence is classified as a tautology, then it will not make a difference whether we choose one translation key or another for purposes of translation. For example, $p \to (q \to p)$ is a tautology, but the same is true for the result of the systematic replacement of its propositional variables for different ones, e.g., $r \to (p \to r)$. Nor will it make a difference to translate 'unless' in terms of 'if not' or to translate it in terms of 'or' as the appropriate translations will be equivalent to each other, and one will be a tautology or a contradiction or neither if, and only if, the other is.

Structural ambiguity presents a separate problem of its own, since it provides us with cases in which one and the same English sentence appears to admit more than one translation into propositional logic, and there is no reason to expect them to be equivalent to each other. In such cases, it would be better to speak of translations of different *interpretations* of the original sentence. In what follows, we will set the issue of ambiguity to one side and continue, for all intents and purposes, to talk of translation of a sentence.

Natural Language and Propositional Logic

We start with two categories for sentences of natural language.

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Tautology
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Tautology An English sentence is a *tautology* if, and only if, its translation into propositional logic is a tautology.

Example 6.1. Consider the English sentence:

Unless I purchase a lottery ticket, it is not the case that I purchase a lottery ticket and I win the lottery

This sentence is a tautology because it translates into a tautology of propositional logic:

$$\neg p \to \neg(p \wedge q)$$

Translation key:

p: I purchase a lottery ticket.
q: I win the lottery

But the sentence we obtain is a tautology:

p	q	$\neg p \to \neg(p \wedge q)$
T	T	T
T	F	T
F	T	T
F	F	T

A tautology is a *logical truth*, since it is true in virtue of the form it exemplifies. But that is not to suggest that every logical truth of English is a tautology. Some English sentences, e.g., 'all human beings are human', should count as a logical truth in virtue of the form they exemplify, yet they do not translate into a tautology of propositional logic.

On the face of it, the English sentence 'all human beings are human', is a logical truth since it exemplifies the form 'all H are H' all of whose instances appear to be true. Yet, the sentence translates into a propositional letter of the form p in propositional logic.

```
Contradiction
```

Contradiction An English sentence is a *contradiction* if, and only if, it translates into a contradiction of propositional logic.

Example 6.2. Consider the English sentence:

You will not purchase a lottery ticket though it is not true that you will not purchase one or you will win.

This sentence is a contradiction because it can be translated into the language of propositional logic as follows:

$$\neg p \wedge \neg(\neg p \vee q)$$

Translation key:

p: I will purchase a lottery ticket.
q: I will win the lottery

But the sentence we obtain is a contradiction:

p	q	$\neg p \wedge \neg(\neg p \vee q)$
T	T	F
T	F	F
F	T	F
F	F	F

A contradiction is logically false or false in virtue of the form it exemplifies. Note, however, that some logically false sentences of English do not translate into contradictions of propositional logic, e.g., 'some human beings are not human' does not. Part of the problem is that propositional logic is not sufficiently rich to capture the structural aspects of the sentence in virtue of which it is false.

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Propositional Equivalence
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Propositional Equivalence Two English sentences are *propositionally equivalent* if, and only if, they translate into equivalent formulas of propositional logic.

Example 6.3. Consider the two English sentences:

You will not win the lottery unless you purchase a ticket.

You will win the lottery only if you purchase a ticket.

These sentences are propositionally equivalent because they translate into two equivalent sentences of propositional logic:

$$\neg p \to \neg q$$
$$q \to p$$

Translation key:

p: I will purchase a lottery ticket.
q: I will win the lottery

p	q	$\neg p \to \neg q$	$q \to p$
T	T	T	T
T	F	T	T
F	T	F	F
F	F	T	T

If two natural language sentences are propositionally equivalent, then they are logically equivalent. On the other hand, two sentences of English may be logically equivalent even if they are not propositionally equivalent. The sentences 'Not all logicians are philosophers' and 'Some logicians are not philosophers' entail each other because they exemplify the forms 'Not all L are P' and 'Some L are not P'. So, they are presumably logically equivalent, even though they are not propositionally equivalent; their translations into propositional logic are not equivalent.

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Propositional Consistency
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Propositional Consistency A set of English sentences is *propositionally consistent* if, and only if, they translate into the members of a consistent set of formulas of propositional logic.

Example 6.4. Consider the set of three English sentences:

You will not win the lottery unless you purchase a ticket.
You will purchase a ticket.
You will not win the lottery.

The set of three sentences is propositionally consistent because the set of its translations into propositional logic is consistent:

$$\neg p \to \neg q$$
$$p$$
$$\neg q$$

Translation key:

p: I will purchase a lottery ticket.
q: I will win the lottery

To check that the set is consistent, we construct a truth table in search of an assignment making true all three formulas.

p	q	$\neg p \to \neg q$	p	$\neg q$
T	T	T	T	F
T	F	T	T	T
F	T	F	F	F
F	F	T	F	T

```
Propositional Validity
```

Propositional Validity An argument in English is *propositionally valid* if, and only if, it translates into a valid argument of propositional logic.

Example 6.5. Consider the natural language argument:

> You will be rich tomorrow if you buy a lottery ticket from me. You will buy a ticket from me if the lottery is rigged and you think that I'm part of the scheme. The lottery is rigged but you will not be rich tomorrow. So, you do not think that I'm part of the scheme.

In premise-conclusion form:

1. You will be rich tomorrow if you buy a lottery ticket from me.
2. You will buy a ticket from me if the lottery is rigged and you think that I'm part of the scheme.
3. The lottery is rigged but you will not be rich tomorrow.
4. So, you do not think that I'm part of the scheme.

Here is the translation of the argument into the language of propositional logic.

1. $p \to q$
2. $(r \land s) \to p$
3. $r \land \neg q$
4. $\neg s$

Translation key:

p: You will buy a lottery ticket from me.
q: You will be rich tomorrow.
r: The lottery is rigged.
s: You think I'm part of the scheme.

We now use a natural deduction proof in order to justify the validity of the argument formulated in propositional logic:

The translation of the argument into propositional logic is valid, which makes the original argument propositionally valid.

To be sure, propositional validity is a species of validity, but not every valid argument is propositionally valid. Some valid arguments translate into invalid arguments of propositional logic simply because propositional logic is not well equipped to capture the structural aspects of the argument that are responsible for its validity in English. Let us look at an example of a putative valid argument that is *not* propositionally valid.

Example 6.6. Consider the natural language argument:

> Unless the mind is material, each human being has an immaterial part. Descartes is a human being. Therefore, unless the mind is material, Descartes has an immaterial part.

In premise-conclusion form, the argument would read:

1. Unless the mind is material, every human being has an immaterial part.

2. Descartes is a human being.
3. Descartes has an immaterial part.

Here is the translation of the argument into the language of propositional logic.

1. $\neg p \to q$
2. r
3. $\neg p \to s$

Translation key:

p: The mind is material.
q: Each human being has an immaterial part.
r: Descartes is a human being. s: Descartes has an immaterial part.

But it is not difficult to verify that the argument formulated in the language of propositional logic is invalid. Here is an assignment on which the premises come out true and the conclusion false:

p	q	r	s	$\neg p \to q$	r	$\neg p \to s$
F	T	T	F	T	T	F

And yet, a closer look at the argument formulated in English suggests that it exemplifies a valid argument form, just not one that is captured in the language of propositional logic:

1. Unless p, every H is I.
2. D is H
3. Unless p, D is I

Expressive Limitations

A variety of valid arguments in English fail to be propositionally valid. That is, the arguments appear exemplify a valid argument form, but their translation into the language of propositional logic is not valid. For a particularly simple example, consider the argument:

> All logicians are philosophers. George is a logician. Therefore, George is a philosopher.

The argument is not propositionally valid because both premises and conclusion translate into propositional variables; they are not constructed from simpler sentences by means of a connective. But the reason the argument is valid is that the sentences display a measure of complexity, one that is not captured by propositional logic. What is required for present purposes is a more powerful and sophisticated formal language, one that does justice to the structure of the three sentences and the argument form they exemplify.

In order to do better, we have to find a way discern more structure in sentences such as ‘Peter is a logician’ and ‘Peter is a philosopher’. One observation is that we can discern constituents of two broad types in each sentence. They each contain a word, i.e., ‘George’, whose primary semantic function is to *designate* an individual, And they each contain an expression, i.e., ‘is a logician’ and ‘is a philosopher’, whose primary semantic function is to *predicate* something of an individual. Some of the sentences we treated as simple in the context of propositional logic may now be regarded as complex sentences, which combine one or more *designators* with a *predicate*.

`Predicates and Designators`

Predicates and Designators The primary semantic function of a *designator* is to designate an individual, and the primary semantic function of a *predicate* is to predicate something of an individual.

Example 6.7. Each of the sentences below combines one or more designators with a predicate:

$$\underbrace{\text{Los Angeles}}_{\text{designator}}\ \underbrace{\text{is a city}}_{\text{predicate}}.$$

$$\underbrace{\text{Los Angeles}}_{\text{designator}}\ \underbrace{\text{is south of}}_{\text{predicate}}\ \underbrace{\text{San Francisco}}_{\text{designator}}$$

$$\underbrace{\text{George}}_{\text{designator}}\ \underbrace{\text{is a philosopher}}_{\text{predicate}}.$$

$$\underbrace{\text{I}}_{\text{designator}}\ \underbrace{\text{talked to}}_{\text{first part of the predicate}}\ \underbrace{\text{her}}_{\text{designator}}\ \underbrace{\text{over zoom}}_{\text{second part of the predicate}}$$

Notice that while some predicate expressions, e.g., ‘... is a philosopher’, combine with just one designator in order to form a sentence, others, e.g., ‘... is south of ...’ or ‘comes between ... and ...’ require more than one designator. Once we acknowledge the distinction between designators and predicates, we realize that the methods of propositional logic are much too crude to represent the structure of predication.

There is another category of expression we find in simple sentences that translate into propositional variables of propositional logic. There is, in particular, an important difference between the sentences ‘George is a logician’ and ‘Someone is a logician’. The two sentences contain a predicate expression, i.e., ‘... is a logician’, which at least in one case combines with a designator, i.e., ‘George’. The crucial observation now is that the word ‘Someone’ is *not* a designator, since its primary semantic function is *not* to designate an individual. For what individual could that be? ‘Someone’ does not designate Peter or any other logician for that matter.

The contrast between quantification and designation is perhaps more clear in the case of quantifier expressions such as ‘no one’ as it occurs in the sentence ‘No one is a logician’. Whatever the semantic function of the quantifier expression ‘no one’, it is *not* to designate an individual of which something is predicated. In fact, the truth conditions of the sentence require the existence of no logicians at all.

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Quantifiers
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Quantifiers The primary semantic function of a quantifier is to express generality.

Some quantifier expressions combine a quantifier, which is a mark of generality, with a predicate in order to form a quantifier phrase. The combination, for example, of the quantifier ‘some’ with the predicate ‘book’ results in the quantifier phrase: ‘some book’. Likewise for the combination of a quantifier such as ‘most’ with a predicate such as ‘material object’, which results in the quantifier phrase: ‘most books’.

Example 6.8. Each of the sentences below combine occurrences of designators, predicates, and quantifier expressions:

$$\underbrace{\text{Something}}_{\text{quantifier}}\ \underbrace{\text{is a city}}_{\text{predicate}}.$$

$$\underbrace{\text{Los Angeles}}_{\text{designator}}\ \underbrace{\text{is south of}}_{\text{predicate}}\ \underbrace{\text{some city}}_{\text{quantifier phrase}}$$

$$\underbrace{\text{Every logician}}_{\text{quantifier phrase}}\ \underbrace{\text{is a philosopher}}_{\text{predicate}}.$$

$$\underbrace{\text{A workday}}_{\text{quantifier phrase}}\ \underbrace{\text{comes between}}_{\text{first part of the predicate}}\ \underbrace{\text{Tuesday}}_{\text{designator}}\ \underbrace{\text{and}}_{\text{second part of the predicate}}\ \underbrace{\text{Thursday}}_{\text{designator}}$$

$$\underbrace{\text{Someone}}_{\text{quantifier}}\ \underbrace{\text{talked to}}_{\text{first part of the predicate}}\ \underbrace{\text{her}}_{\text{designator}}\ \underbrace{\text{over zoom}}_{\text{second part of the predicate}}$$

Exercises

1. Determine whether the following English sentences are tautologies, contradictions, or neither tautologies nor contradictions:

   a. Someone is a logician and not a logician.
   b. Some logicians are philosophers but some logicians are not philosophers.
   c. If all logicians are philosophers, then if some philosophers are logicians, then all logicians are philosophers.
   d. Someone is a logician and someone is not a logician.
   e. All logicians are philosophers but no philosophers are logicians.

2. Determine whether the following arguments are propositionally valid:

a. The deal will be completed only if Smith or Jones are in attendance. If Smith attends, then Jones will too. Jones will not be in attendance. Therefore, the deal will not be completed.
b. All logicians are philosophers. All logicians learn propositional logic at some point. So, some philosophers learn propositional logic at some point.
c. The deal will not be completed unless someone is in attendance. Neither Smith nor Jones will be in attendance. Therefore, the deal will not be completed.

3. Identify occurrences of designators, predicates, and quantifier expressions in the following sentences:

a. USC is located in Los Angeles.
b. Some private university is located in NYC.
c. Some critic admires every critic.
d. Mark Twain wrote *Tom Sawyer*.
e. Every student owns a computer.
f. Everyone admires someone.
g. Some workday comes after a weekend day.

Part III

Quantificational Logic

7

Formal Language

The language of quantificational logic will afford the means to capture the validity of a variety of arguments for which propositional logic appears to be inadequate. We specify:

- a *syntax* for the language, which will include a vocabulary and a set of grammatical rules designed to specify which expressions are formulas of the language, and
- a *semantics* for the language, which will explain how to interpret the expressions and formulas of the language and define what is for a formula to be true under a given interpretation.

Syntax

We begin with the vocabulary of quantificational logic.

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Vocabulary
```

The vocabulary of quantificational logic contains six types of symbol:

Constants These are the lowercase letters a, b, c, d, and e with or without numerical subscripts:

$$a, b, c, d, e$$

Predicates These are uppercase letters P, Q, R, S, and T with or without numerical subscripts. Each predicate comes with a fixed number of argument places.

$$P, Q, R, S, T \dots$$

Variables These are lowercase letters x, y, and z with or without numerical subscripts:

$$x, y, z \ldots$$

Connectives These are the symbols:

$$\neg, \vee, \wedge, \rightarrow$$

Quantifiers There are two quantifier expressions:

$$\forall, \exists$$

Parentheses There are two parentheses:

$$), ($$

Nothing else is a symbol of the language.

`Grammar`

The grammar of quantificational logic explains how to combine these symbols into formulas of the language. We proceed in two stages.

Atomic Formula If P is a predicate with n argument places, and each of $\tau_1, \ldots, \tau_n$ is a constant or a variable, then

$$P\tau_1 \ldots \tau_n$$

is an *atomic formula*.

Atomic formulas are the formal counterparts of simple predications such as 'Los Angeles is a city' or 'Los Angeles is between San Diego and San Francisco'.

Example 7.1. Each of the expressions below is an atomic formula of quantificational logic:

$$\begin{gathered} Pxy \\ Qabx \\ Rxaby_3 \\ \ldots \end{gathered}$$

Atomic formulas provide simple constituents for more complex formulas.

Formula We now define what is for an expression to be a formula of quantificational logic.

1. All atomic formulas are *formulas*.

2. If φ and ψ are *formulas*, then each of

$$\neg\varphi, (\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \rightarrow \psi)$$

   is a *formula*.

3. If φ is a formula and x is a variable, then each of $\forall x\varphi$ and $\exists x\varphi$ is a formula.

4. Nothing else is a *formula*.

We proceed to illustrate the characterization of formula through concrete examples.

```
<img src="images/ex-6.2.png">

The construction tree depicts the construction of the formula from simpler constituents.
```

Example 7.2. The expression below is a formula of quantificational logic:

$$(\forall x Pxy \rightarrow \exists y\ Qab)$$

By rule 2, $(\forall x Pxy \rightarrow \exists y\ Qab)$ is a formula if $\forall x\ Pxy$ and $\exists y\ Qab$ are each a formula.
By rule 3, $\forall x\ Pxy$ is a formula if Pxy is a formula.
By rule 3, $\exists y\ Qab$ is a formula if Qab is a formula.
By rule 1, each Pxy and Qab are formulas, since they are each atomic formulas.

Therefore, we conclude that $(\forall x Pxy \rightarrow \exists y\ Qab)$ is a formula of quantificational logic.

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<img src="images/ex-6.3.png">

The construction tree depicts the construction of the formula from simpler constituents.
```

Example 7.3. The expression below is a formula of quantificational logic:

$$\forall x(Pxy \rightarrow Qab)$$

By rule 3, $\forall x(Pxy \rightarrow Qab)$ is a formula if $(Pxy \rightarrow Qab)$ is a formula.
By rule 2, $(Pxy \rightarrow Qab)$ is a formula if each Pxy and Qab are formulas.
By rule 1, each Pxy and Qab are formulas, since they are each atomic formulas.

Therefore, we conclude that $\forall x(Pxy \rightarrow Qab)$ is a formula of quantificational logic.

We now make a distinction between two types of occurrences of a variable in a formula. In the formula $\forall x(Qxy \to Ryx)$ the last two occurrences of the variable x match the variable accompanying the quantifier expression, whereas the occurrences of the variable y do not. The occurrences of the variable x have been captured by the initial quantifier, but the occurrences of the variable y remain free.

`Free Occurrences`

Free Occurrence of a Variable We define what is for an occurrence of a variable to be *free* in a formula:

1. All occurrences of a variable in an atomic formula are free.
2. The free occurrences of a variable in formulas of the form φ and ψ remain free when they occur in

$$\neg\varphi, (\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \to \psi)$$

3. No occurrence of the variable v is free in a formula of the form

$$\forall v\varphi, \exists v\varphi$$

All occurrences of variables other than v that are free in φ remain free in those formulas.

Example 7.4. The first two occurrences of the variable x occur free in the formula:

$$Px \to (Qxy \to \forall xRxx)$$

By rule 1, x occurs free in Px, which is an atomic formula, and, by rule 3, the occurrence remains free when it occurs in a conditional of the form $Px \to \psi$.

By rule 1, x occurs free in Qxy, which is an atomic formula, and, by rule 2, the occurrence remains free when it occurs in a formula of the form $(Qxy \to \psi)$.

By rule 2, the last two occurrences of x in $\forall xRxx$ are not free.

An occurrence of a variable is free when it has not been captured by a quantifier. In the last example, the last two occurrences of the variable x are under the scope of the universal quantifier $\forall x$, whereas the first two occurrences of the variable are not under the scope of a quantifier.

`Bound Occurrences`

Bound Occurrence of a Variable: An occurrence of a variable in a formula is *bound* if, and only if, it is not free in that formula.

A variable occurs *freely* in a formula if, and only if, some of its occurrences in the formula are *free*.

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Open and Closed Formulas
```

Open and Closed Formulas A formula is *open* if, and only if, some variables occur freely in the formula. Otherwise, the formula is *closed*.

Example 7.5. The formula below is open:

$$(\forall x \exists y Rxy \vee Qyx)$$

This is because the last occurrences of the variables y and x are free in the formula.

Example 7.6. The formula below is closed:

$$\forall x \exists y (Rxy \vee Qyx)$$

The difference with respect to the last formula is that all occurrences of the variables y and x are now captured by the quantifiers $\exists y$ and $\forall x$, respectively.

Notational Convention We may remove the outer parentheses from a formula that is not part of another formula.

Notice that this convention will *not* allow us to omit the parentheses in a formula such as:

$$\forall x (Rxa \rightarrow Rax).$$

For $(Rxa \rightarrow Rax)$ is here part of the formula $\forall x(Rxa \rightarrow Rax)$. Such a formula is importantly different from:

$$(\forall x Rxa \rightarrow Rax).$$

The former formula is a *closed formula* whereas the latter is an *open formula*, since the last occurrence of the variable x is free in $(\forall x Rxa \rightarrow Rax)$ as it is not in the scope of the universal quantifier $\forall x$.

Example 7.7. We are able to use the expression

$$\forall x Rxa \rightarrow Rax$$

as an abbreviation for the formula

$$(\forall x Rxa \rightarrow Rax)$$

Exercises

1. Determine whether each of the following expressions is an atomic formula of quantificational logic.

   a. $\neg Rxy$

b. Rxy

c. $Aabx$

d. Qe

e. $Pxyza$

f. RPa

2. Determine whether each of the following expressions is a formula of quantificational logic.

a. $\neg\neg Rxy$

b. $(Pa \to (Axy \wedge \exists x Qx))$

c. $\forall x Ryy$

d. $\exists xy(P \wedge P(y))$

e. $\forall\forall x Ryx$

f. $\exists x \exists y Rxy$

3. Determine whether each of the following expressions is an *abbreviation* for a formula of quantificational logic.

a. $\exists x \exists y Rxy \to Px$

b. $\forall x Rx \vee \exists y Qyy$

c. $Px \wedge Qx \vee Rxy$

d. $Px \wedge Qx \to Rxy$

e. $\exists x Px \wedge \exists y Qy \wedge \exists z Rz$

f. $\exists x \forall y \exists z(Px \wedge Qy \wedge Rxyz)$

4. Determine whether the following formulas contain a free occurrence of the variable x.

a. $\forall x(Rxy \wedge Ryx)$

b. $\forall x Rxy \wedge Ryx$

c. $\exists x(Px \wedge Qx) \to \forall y Rxy$

d. $\forall x \exists y Rxy \to \exists x \forall y Ryx$

e. $\forall x \exists y Rxy \to \forall y Ryx$

f. $\forall x(\exists y Rxy \to \forall y Ryx)$

5. Determine whether the following formulas are *open* or *closed*. Justify your answers.

a. $\forall x Rxy \wedge Ryx$

b. Rab

c. $\exists x(Pax \wedge Qxa) \to \forall y Rxy$

d. $\forall x Ryy$

e. $\forall x \exists y \forall z Rxyx$

f. $Rax \to \forall x Rxx$

8

Semantics

We now explain how to interpret the language of quantificational logic. We will not use assignments of truth values to interpret the language but rather models, which will assign a semantic value to the simplest constituents of a formula. A model for quantificational logic must specify:

1. a *domain* of discourse for the quantifiers to range over,
2. *denotations* for the constants of the language, and
3. *extensions* for the predicates of the language

The domain of discourse determines the truth conditions of quantified formulas in much the way in which the truth value of a natural language sentence such as 'I packed everything into my suitcase' depends on what is the domain of objects over which we take the quantifier 'everything' to range. In what follows, we will use a non-empty set of objects as the domain of a model. There are no restrictions on the domain of discourse other than it must contain at least one object.

We will let a model specify denotations for the constants of the language. Constants are formal counterparts of names in natural language, which are generally taken to denote an object in the domain of discourse, e.g., 'Los Angeles' denotes a city in Southern California, and 'Silver Lake' denotes an area of Los Angeles. A model will assign a member of the domain to each constant of the language as a denotation.

The case of predicates is more subtle. Predicates are formal counterparts of predicate expressions in natural language, which are true of some objects and not of others. Let us begin with one-place predicates such as 'is red', which applies to red objects and does not apply to non-red objects. One suggestion at this point is to identify the interpretation of 'red' with the *set* of objects to which the predicate applies. The set of objects to which a given predicate applies is known as the *extension* of the predicate. Thus the extension of the predicate ‘red’ is the set of red objects and the extension of the predicate ‘human being’ is the set of human beings.

The case of two-place predicates is more complicated. The predicate ‘is south of’ applies to Los Angeles and San Francisco *in that order* but it fails to apply to San Francisco and Los Angeles *in that order*. This raises the question of how to specify the extension of the predicate. The answer is to use sets of *ordered pairs* of objects in the domain, e.g., $\langle LA, SF\rangle$ is the ordered pair of Los Angeles and San Francisco in that order whereas $\langle SF, LA\rangle$ is the ordered pair of San Francisco and Los Angeles in that order. The extension of the two-place predicate ‘is south of’ will be a set of ordered pairs, which contains $\langle LA, SF\rangle$ but not $\langle SF, LA\rangle$. The solution generalizes to predicates with even more argument places. The interpretation of a three-place predicate is a set of ordered triples, and more generally, the interpretation of an n-place predicate is a set of ordered n-tuples.

Models

Given an argument in the language of quantificational logic, we will often consider the question of whether there is an interpretation of the language on which the premises are true and the conclusion false. That amounts to the question whether there is a model for quantificational logic, which verifies the premises but not the conclusion of the argument.

```
Model
```

Model A model M for quantificational logic is an ordered pair $\langle D, I\rangle$, where:

- D is a non-empty set of objects, and
- I interprets constants and predicates in accordance to the requirements:
  - If a is a constant, $I(a)$ is a member of D.
  - If P is a one-place predicate, $I(P)$ is a set of members of P.
  - If R is an n-place predicate, $I(R)$ is a set of n-tuples of members of D.

Example 8.1. Consider the argument given below.

1. $(Pa \wedge Qb) \rightarrow Rab$
2. $\neg Rba$
3. $\exists x(Px \wedge Rxa)$

We may evaluate the premises and conclusion of the argument in a model M given by $\langle D, I\rangle$, where:

- $D = \{1, 2\}$
- $I(a) = 1$
- $I(b) = 2$
- $I(P) = \{1, 2\}$
- $I(Q) = \{2\}$
- $I(R) = \{\langle 1, 2\rangle\}$

We will eventually explain that the model verifies the premises but not the conclusion of the argument.

One heuristic you may find helpful for the specification of a model is to use a diagrams as partial representation of the model. The diagram should come accompanied with a key designed to help us specify the interpretation of the relevant constants and predicates.

```
<img src="images/ex 6.8.png" style="display:block; margin: 0 auto;">\
$P :  \text{enclosed by a circle}$\
$Q : \text{enclosed by a dark grey figure}$\
$R : \text{points to}$\
$a : p_1$\
$b : p_2$
```

Example 8.2. The diagram included in the margin encodes the partial specification of a model $\langle D, I\rangle$, where:

- $D = \{p_1, p_2\}$
- $I(a) = p_1$
- $I(b) = p_2$
- $I(P) =$ enclosed by a circle
- $I(Q) =$ enclosed by a dark grey figure
- $I(R) =$ points to

Notice that the diagram comes with a key designed to explain how to interpret each predicate and constant in the diagram.

This heuristic comes with severe limitations, since some models, e.g., models with an infinite domain, are much too complex to be represented by means of a finite diagram. The method will, however, suffice for *some* purposes; it will enable us to establish the invalidity of a broad family of arguments in quantificational logic.

Truth Relative to a Variable Assignment

The definition of a model makes sure that models provide a semantic value for constants and predicates, but they remain silent when it comes to variables. If constants are the formal counterparts of names in natural language, variables resemble pronouns like ‘he’ or ‘she’, which do not appear to receive a denotation when used in a sentence like ‘If a person likes opera, then she likes classical music.' For this reason, models do not assign a denotation to variables.

Instead, variables will be treated as temporary labels for members of the domain. That will allow us to evaluate an open formula $Px \wedge Rxy$ relative to an assignment of values to the variables that occur in it. More generally, given a model M, we now explain what is an assignment of members of M to the variables of the language.

Variable Assignment A *variable assignment* α over a model M is a map from the range of variables to a members of the domain D.

Example 8.3. Each of the maps depicted below are variable assignments over a model M with domain $\{p_1, p_2\}$:

x	y	z	x_1	y_1	z_1	x_2	y_2	z_2	$\cdots$
p_1	p_2	p_1	p_2	p_1	p_2	p_1	p_1	p_2	$\cdots$

x	y	z	x_1	y_1	z_1	x_2	y_2	z_2	$\cdots$
p_2	p_1	p_2	p_1	p_2	p_1	p_2	p_1	p_2	$\cdots$

x	y	z	x_1	y_1	z_1	x_2	y_2	z_2	$\cdots$
p_1	p_2	p_2	p_2	p_2	p_1	p_2	p_1	p_2	$\cdots$

Once variable assignments are in place, we are in a position to assign a semantic value to a variable x in a model M *relative to an assignment* α, which is just $\alpha(x)$.

For technical reasons, we want to assign semantic values to open formulas in a model relative to a variable assignment over the model. The definition of truth in a model relative to a variable assignment will be inductive: we will define the truth conditions of complex formulas in terms of the truth conditions of simpler ones in much the way with did with the definition of truth under an assignment of truth values to propositional variables in the case of propositional logic. We will begin with an account of the truth conditions of atomic formulas, and we will explain how to assign truth values to more complex sentences in a model relative to an assignment.

Denotation If τ is either a constant c or a variable x, the denotation of τ in a model M under assignment α is $I(c)$ if τ is a constant c and $\alpha(x)$ if τ is a variable x.

Atomic Formulas $P\tau$ is true in M under α if the denotation of τ in M under α is in the extension of P, e.g., Px is true in M under α if $\alpha(x) \in I(P)$ and Pc is true in M under α if $I(c) \in I(P)$. More generally, $R\tau_1 \cdots \tau_n$ is true in M under α if the denotations of $\tau_1 \cdots \tau_n$ are in that order in the extension of R, e.g., Rax is true in M under α if $\langle I(a), \alpha(x)\rangle \in I(R)$.

Example 8.4.

```
$D = \{1, 2\}$\
$I(P) = \{1, 2\}$\
$I(Q) = \{2\}$\
$I(R) = \{\langle 1, 2\rangle\}$\
$I(a) = 1$\
$I(b) = 2$
```

Let α be an assignment for the model M:

x	y	z	x_1	y_1	z_1	...
1	2	2	2	2	1	...

Then:

The formula Rxy is true in M relative to α because $\langle\alpha(x), \alpha(y)\rangle = \langle 1, 2\rangle$, which is in $I(R)$.

The formula Ray is true in M relative to α because $\langle I(a), \alpha(y)\rangle = \langle 1, 2\rangle$, which is in $I(R)$.

The formula Px_1 is true in M relative to α because $\alpha(x_1) \in I(P)$, that is, $2 \in I(P)$.

The evaluation of atomic formulas is similar when we use a diagram to specify a model.

```
<img src="images/ex 6.8.png" style="display:block; margin: 0 auto;">\
$P :  \text{enclosed by a circle}$\
$Q : \text{enclosed by a dark grey figure}$\
$R : \text{points to}$\
$a : p_1$\
$b : p_2$
```

Example 8.5. Consider a model M depicted by the diagram, and let α be an assignment of the form:

x	y	z	x_1	y_1	z_1	...
p_1	p_2	p_2	p_2	p_2	p_1	...

Then:

The formula Rxy is true in M relative to α because $\alpha(x)$ points to $\alpha(y)$. That is, p_1 points to p_2.

The formula Ray is true in M relative to α because p_1 points to $\alpha(y)$. That is, p_1 points to p_2.

The formula Px_1 is true in M relative to α because $\alpha(x_1)$, that is, p_2, is enclosed by a circle.

If a complex formula is constructed from simpler formulas by an application of a connective such as $\neg$, $\wedge$. $\vee$, $\rightarrow$, and $\leftrightarrow$, then its truth conditions in a model M relative to an assignment α are given inductively in the style of the definition of truth for propositional logic. That is,

Negation $\neg\varphi$ is true in M relative to a variable asignment α if, and only if, φ is not true in M relative to α.

Conjunction $\varphi \wedge \psi$ is true in M relative to a variable asignment α if, and only if, φ is true in M relative to α and ψ is true in M relative to α.

Disjunction $\varphi \vee \psi$ is true in M relative to a variable asignment α if, and only if, φ is true in M relative to α or ψ is true in M relative to α.

Conditional $\varphi \rightarrow \psi$ is true in M relative to a variable asignment α if, and only if, φ is not true in M relative to α or ψ is true in M relative to α.

Example 8.6.

```
$D = \{1, 2\}$\
$I(P) = \{1, 2\}$\
$I(Q) = \{2\}$\
$I(R) = \{\langle 1, 2\rangle\}$\
$I(a) = 1$\
$I(b) = 2$
```

Let α be an assignment for the model M:

x	y	z	x_1	y_1	z_1	...
1	2	2	2	2	1	...

Then:

$Pa \wedge Qb$ is true in M relative to α because Pa is true in M relative to α and Qb is true in M relative to α. The former is true because $p_1 \in I(P)$ and the latter is true because $p_2 \in I(Q)$.

$(Pa \wedge Qb) \rightarrow Rab$ is true in M relative to α because Rab is true in M relative to α. This is because $\langle I(a), I(b)\rangle = \langle p_1, p_2\rangle \in I(R)$.

$\neg Rba$ is in M relative to α because Rba is *not* true in M relative to α. This is because $\langle I(b), I(a)\rangle = \langle p_2, p_1\rangle \notin I(R)$.

The situation is similar for diagramatic representations of a model:

```
<img src="images/ex 6.8.png" style="display:block; margin: 0 auto;">\
$P :  \text{enclosed by a circle}$\
$Q : \text{enclosed by a dark grey figure}$\
$R : \text{points to}$\
$a : p_1$\
$b : p_2$
```

Example 8.7. Consider a model M depicted by the diagram, and let α be an assignment of the form:

x	y	z	x_1	y_1	z_1	...
p_1	p_2	p_2	p_2	p_2	p_1	...

Then:

$Pa \wedge Qb$ is true in M relative to α because Pa is true in M relative to α and Qb is true in M relative to α. The former is true because p_1 is enclosed by a circle and the latter is true because p_2 is enclosed by a grey figure.

$(Pa \wedge Qb) \rightarrow Rab$ is true in M relative to α because Rab is true in M relative to α. This is because $I(a)$ points to $I(b)$. That is, p_1 points to $p2$.

$\neg Rba$ is in M relative to α because Rba is *not* true in M relative to α. This is because $I(b)$ does not point to $I(b)$. That is, p_2 does not point to $p1$.

We now explain how to evaluate quantified formulas of the form $\forall v\varphi$ and $\exists v\varphi$ in a model M relative to a variable assignment over M. We want a formula of the form $\exists xP(x)$ to be true in M relative to α if some member of the domain of M lies in the extension of P. One way to achieve this is to declare the formula true in M relative to α if there is an object in the domain of M that we can assign to x in order to verify $P(x)$, which means that there is an assignment β which is just like α except perhaps for the fact that it assigns a different object to x such that $P(x)$ is true in M relative to β. Similarly, we want a formula $\forall xP(x)$ to be true in M relative to α if every member of the domain of M lies in the extension of P. In other words, $P(x)$ is true in M no matter what we assign to x provided the new variable assignment remains just like α in all other respects.

Existential Quantification $\exists v\varphi$ is true in M relative to a variable assignment α if, and only if, there is an object p in the domain of M such that φ is true in M relative to $\alpha[x/p]$, which is an assignment just like α except perhaps for the fact that it assigns p to the variable x.

Universal Quantification $\forall v\varphi$ is true in M relative to a variable assignment α if, and only if, for every object p in the domain of M, φ is true in M relative to $\alpha[x/p]$, which is an assignment just like α except perhaps for the fact that it assigns p to the variable x.

Example 8.8.

```
$D = \{1, 2\}$\
$I(P) = \{1, 2\}$\
$I(Q) = \{2\}$\
$I(R) = \{\langle 1, 2\rangle\}$\
$I(a) = 1$\
$I(b) = 2$
```

Let α be an assignment for the model M:

x	y	z	x_1	y_1	z_1	...
1	2	2	2	2	1	...

Then:

$\exists x(Px \wedge \neg Rxa)$ is not true in M relative to α because there is no member of D we can assign to the variable x, where the new variable assignment remains just like α in all other respects, in order to make $Px \wedge \neg Rxa$ true:

- $\alpha[x/1]$: $Px \wedge \neg Rxa$ is not true relative to that assignment because $\langle 1, I(a)\rangle \notin I(R)$.
- $\alpha[x/2]$: $Px \wedge \neg Rxa$ is not true relative to that assignment because $\langle 2, I(a)\rangle \notin I(R)$.

$\forall xPx$ is true in M relative to α because no matter what member of D assign to the variable x, where the new variable assignment remains just like α in all other respects, we find that Px is true in M relative to the new assignment:

- $\alpha[x/1]$: Px is true relative to that assignment because $1 \in I(P)$.
- $\alpha[x/2]$: Px is true relative to that assignment because $2 \in I(P)$.

The situation is similar for diagramatic representations of a model M relative to α because there is no member of D we can assign to the variable x making sure the new variable assignment remains just like α in all other respects in order to make:

```
<img src="images/ex 6.8.png" style="display:block; margin: 0 auto;">\
$P :  \text{enclosed by a circle}$\
$Q : \text{enclosed by a dark grey figure}$\
$R : \text{points to}$\
```

```
$a : p_1$\
$b : p_2$
```

Example 8.9. Consider a model M depicted by the diagram, and let α be an assignment of the form:

x	y	z	x_1	y_1	z_1	...
p_1	p_2	p_2	p_2	p_2	p_1	...

Then:

$\exists x(Px \wedge \neg Rxa)$ is not true in M relative to α because there is no member of D we can assign to the variable x making sure the new variable assignment remains just like α in all other respects in order to make: $Px \wedge \neg Rxa$:

- $\alpha[x/p_1]$: $Px \wedge \neg Rxa$ is not true relative to that assignment because p_1 does not point to p_1.
- $\alpha[x/p_2]$: $Px \wedge \neg Rxa$ is not true relative to that assignment because p_2 does not point to p_1.

$\forall x Px$ is true in M relative to α because no matter what member of D assign to the variable x, where the new variable assignment remains just like α in all other respects, we find that Px is true in M relative to the new assignment:

- $\alpha[x/p_1]$: Px is true relative to that assignment because p_1 is enclosed by a circle.
- $\alpha[x/p_2]$: Px is true relative to that assignment because p_2 is enclosed by a circle.

Truth in a Model

Once we define what is for a formula φ to be true in a model M relative to an assignment α, we are in a position to define what is for a formula φ to be true in a model M.

Truth in a Model A formula φ is *true in a model* M if, and only if, φ is true in M relative to *all* variable assignments over M.

Logical Truths

Logical Truth A formula φ of quantificational logic is a *logical truth* if, and only if, φ is true in every model.

One difference between propositional logic and quantificational logic is that we are not in a position to survey all interpretations of the language in order to determine whether a formula is a logical truth. There is an infinity of domains of discourse to consider and there are different interpretations available for the constants and predicates of the language available for each such domain. The generalization of the tableaux method will provide us with an indirect method for establishing the logical truth of a formula, but in the meantime, all we are in a position to do is to find a counterexample model when a formula is *not* a logical truth.

```
<img src="images/ex 6.19.png" style="display:block; margin: 0 auto;">\

$R : \text{points to}$\
$a : p_1$
```

Example 8.10. The formula $Raa \rightarrow \forall x Rxa$ is *not* a logical truth.

The model depicted by the diagram verifies the antecedent of the conditional Raa but it falsifies the consequent $\forall x Rax$.

The atomic formula Raa is true in the model because p_1 points to itself.

$\forall x Rax$ is not true in the model because Rax does *not* remain true no matter what value we assign to x, e.g., Rax is not true relative to an assignment on which x denotes p_2.

Equivalence and Consistency

Equivalence Two formulas φ and ψ of quantificational logic are *equivalent* if, and only if, they are true in exactly the same models.

We find ourselves in a similar situation. We are not yet in a position to establish that two formulas are logically equivalent, but if they are not, we may set out to find a counterexample in the form of a model in which one formula is true and the other one is false.

```
<img src="images/ex-6.20.png" style="display:block; margin: 0 auto;">\

$P: \text{enclosed by a circle}$ \
$Q: \text{enclosed by a diamond}$
```

Example 8.11. The formulas $\exists x Px \wedge \exists x Qx$ and $\exists x(Px \wedge Qx)$ are *not* a logically equivalent.

The model M depicted by the diagram verifies $\exists xPx \wedge \exists xQx$ but it falsifies $\exists x(Px \wedge Qx)$.

The conjunction $\exists xPx \wedge \exists xQx$ is true in the model because each conjunct is true in the model. $\exists xPx$ is true in the model because Px is true when p_1 is assigned to x, and $\exists xQx$ is true in the model because Qx is true when p_2 is assigned to x.

The existential quantification $\exists x(Px \wedge Qx)$ is not true in the model because the conjunction $Px \wedge Qx$ is never true no matter what we may assign to the variable x.

Consistency A set of formulas S of quantificational logic is *consistent* if, and only if, there is at least one model in which all of its members are true. Otherwise, the set is *inconsistent*.

If a set of formulas is consistent, then we are in a position to search for a model in which all of their members are true, but it is a different matter to conclusively establish that a set of sentences is inconsistent.

```
<img src="images/ex 6.21.png" style="display:block; margin: 0 auto;">\

$P: \text{enclosed by a circle}$ \
$Q: \text{enclosed by a dark grey figure}$\
$R: \text{points to}$\
$a : p_1$\
$b : p_2$
```

Example 8.12. The set of formulas $\{R(a,b), P(a), Q(b), \exists x(P(x) \wedge Q(x)\}$ is consistent.

All these formulas are true in the model depicted by the diagram:

Rab is true in the model because the denotation of a points to that of b. That is, p_1 points to p_2.

Pa is true because the denotation of a, namely, p_1, is enclosed by a circle.

Qb is true because the denotation of b, namely, p_2, is enclosed by a dark grey figure.

$\exists x(Px \wedge Qx)$ is true in the model because the conjunction $Px \wedge Qx$ is true when we assign a member of the domain, namely, p_2, to the variable x.

Validity

Validity An argument of quantificational logic is *valid* if, and only if, there is no model which all of its premises are true while its conclusion is false. Otherwise, the argument is *invalid*.

If an argument is not valid, then we are in a position to search for a model in which the premises are true and the conclusion is false. Otherwise, we will have to rely on a generalization of the tableaux method in order to establish validity.

```
<img src="images/ex 6.8.png" style="display:block; margin: 0 auto;">\

$P :  \text{enclosed by a circle}$\
$Q : \text{enclosed by a dark grey figure}$\
$R : \text{points to}$\
$a : p_1$\
$b : p_2$
```

Example 8.13. Consider the following argument:

1. $(P(a) \wedge Q(b)) \rightarrow R(a,b)$
2. $\neg R(b,a)$
3. $\exists x(P(x) \wedge R(x,a))$

The model depicted by the diagram verifies the premises of the argument but falsifies the conclusion.

The conditional $(Pa \wedge Qb) \rightarrow Rab$ is true in the model because its consequent Rab is true in the model.

The negation $\neg Rba$ is true in the model because the denotation of b, namely, p_2 does not point to the denotation of a, namely, p_1.

The existential quantification $\exists x(Px \wedge Rxa)$ is false in the model because the conjunction $Px \wedge Rxa)$ is never true no matter what we assign to the variable x.

Exercises

1. Determine which of the following formulas are true in the model:

```
<img src="images/prob 6.png" style="display:block; margin: 0 auto;">\

$Q : \text{enclosed by a dark grey figure}$\
$S : \text{enclosed by a circle}$\
$T : \text{enclosed by a diamond}$\
```

```
$R : \text{points to}$\
$a : p_1$\
$b : p_2$
```

```
a. $Rab$

#. $Rab \to Rbb$

#. $Pa \wedge  Qb$

#. $\exists x Sx \wedge \exists x Tx$

#. $\forall x (\exists y  Rxy \vee \exists y Ryx)$

#. $\forall x (Sx  \to \exists y (Qy \wedge Rxy))$
```

9

Translation

We want to use the language of quantificational logic for the evaluation of natural language arguments. That will require us to be able to translate from natural language into the formal language of quantificational logic. We proceed in stages. We will briefly consider the case of predications, which are some of the simplest natural language sentences, and we will pay special attention to the translation of quantifier phrases.

A Translation Method

We build on the translation method we used for propositional logic. That is, we will first indicate how natural language sentences are constructed from simpler sentences by means of the usual propositional connectives. The difference is that we will now look for further structure into the sentences we would have formalized with the help of a propositional variable in propositional logic.

Predication We use predications to translate the most basic parts of the target sentence.

Consider the sentence:

Alice talked to the King, and the King talked to the Queen.

The translation method for propositional logic generates:

(Alice looked at the King $\wedge$ the King looked at the Queen.)

In a further step, we now make use of predicates and constants to translate the simplest sentences:

```
*Translation key*\
$a$: Alice\
$b$: The King\
$c$: The Queen\
$R$ \_  \_ : \_  talked to \_
```

Rab ∧ *Rbc*

Quantifier phrases require more discussion. We will look at the case of simple quantifier expressions such as ‘something’ and ‘everything’ and we will then move to more complex quantifier phrases such as ‘some apple’ and ‘every computer’, which require separate discussion.

Quantifiers

```
We paraphrase the relevant sentences to make clear how they may be built from simpler constituents wi
```

One may rephrase ‘Something is heavy’ as ‘Some *thing* is such that *it* is heavy’. The pronoun ‘it’ will rendered as a variable, which will be bound by the existential quantifier that corresponds to ‘something’. Consider the sentence:

Something is heavy.

We rephrase it with the help of a pronoun, which will eventually be replaced with a variable:

```
*Translation key*\
$P$ \_ : \_ is heavy.
```

Some *thing* is such that *it* is heavy.
$\exists x Px$

Similarly for the sentence:

Everything is a material object.

We now rephrase it with the help of a pronoun, which will eventually be replaced with a variable:

```
*Translation key*\
$Q$ \_ : \_ is a material object.
```

Every *thing* is such that *it* is a material object.
$\forall x Qx$

Similarly for predicates with more than one argument place:

```
*Translation key*\
$a$: San Francisco\
$R$ \_  \_ : \_ is south of \_
```

Something is south of San Francisco
Some *thing* is such that *it* is soutn of San Francisco.
$\exists x Rxa$

Common quantifier phrases Common quantifier phrases combine a quantifier such as 'some' or 'a' and 'all' or 'every' with a monadic predicate such as 'apple' or 'student' as in 'some apples' or 'every city'.

There are two main types of natural language sentence to consider:

Some apples are delicious.
All computers are expensive.

When the quantifier phrase combines 'some' with a monadic predicate, then we proceed as follows:

```
*Translation key*\
$P$ \_ :  \_  is an apple
```

'Some apples ...'
'Some *thing* is such that *it* is an a apple *and* ... *it* ...'
$\exists x(Px \wedge ... x ...)$

Consider the sentence:

Some apples are delicious.

We now rephrase it with the help of a pronoun, which will eventually be replaced with a variable:

```
*Translation key*\
$P$ \_ :  \_  is an apple\
$Q$ \_ : \_ is delicious
```

Some *thing* is such that *it* is an apple and *it* is delicious.
$\exists x(Px \wedge Qx)$

Similarly for the sentence:

```
*Translation key*\
$a$:  San Francisco\
$P$ \_ :  \_ is a city\
$R$ \_ \_ : \_ is south of \_
```

Some city is south of San Francisco
Some *thing* is such that *it* is a city and *it* is south of San Francisco.
$\exists x(Px \wedge Rxa)$

When the quantifier phrase combines 'every' or 'all' with a monadic predicate, then we proceed as follows:

```
*Translation key*\
$P$ \_ : \_ is a computer\
```

Every computer ...
Every *thing* is such that *if it* is a computer, *then* ... *it* ...
$\forall x(Px \to \dots x \dots)$

Consider the sentence:

Every computer is expensive

We now rephrase it with the help of a pronoun, which will eventually be replaced with a variable:

```
*Translation key*\
$P$ \_ :  \_ is a computer\
$Q$ \_ : \_ is expensive
```

Every *thing* is such that if *it* is a computer, then *it* is expensive.
$\forall x(Px \to Qx)$

Similarly for the sentence:

```
*Translation key*\
$P$ \_ : \_  is a computer\
$R$ \_ \_ : \_ is connected to \_ \
$a$: The network
```

Every computer is connected to the network.
Every *thing* is such that if *it* is a computer, then *it* is connected to the network.
$\forall x(Px \to Rxa)$

Multiple Generality Some predicates with multiple argument places may combine with more than one quantifier phrase:

Something is is south of San Francisco
Something is south of something
Some city is south of something
Some city is south of some city

Consider the sentence:

Something is south of something

We now paraphrase the sentence:

```
*Translation key*\
$R$ \_ \_ : \_ is south of \_
```

Some $thing_x$ is such that some $thing_y$ is such that it_x is south of it_y.
$\exists x \exists y Rxy$

Consider now the sentence:

Some city is south of something.

We now paraphrase the sentence:

```
*Translation key*\
$P$ \_ : \_ is a city\
$R$ \_ \_ : \_ is south of \_
```

Some *thing*$_x$ is such that *it*$_x$ is a city and some *thing*$_y$ is such that *it*$_x$ is south of *it*$_y$.
$\exists x(Px \land \exists yRxy)$

For yet another example, consider the sentence:

Some city is south of some city.

We now paraphrase the sentence:

```
*Translation key*\
$P$ \_ :  \_ is a city\
$R$ \_ \_ : \_ is south of \_
```

Some *thing*$_x$ is such that *it*$_x$ is a city and some *thing*$_y$ is such that *it*$_x$ is a city and *it*$_x$ is south of *it*$_y$.
$\exists x(Px \land \exists y(Py \land Rxy))$

Some sentences combine a universal and existential quantifier phrases.

Every student takes some course
Some student takes every course
Some course is taken by every student

Consider the sentence:

Every student takes some course

We now paraphrase the sentence in stages:

```
*Translation key*\
$P$ \_ : \_ is a student\
$Q$ \_ : \_ is a course\
$R$ \_ \_ : \_ takes \_
```

Every *thing*$_x$ is such that if *it*$_x$ is a student, then *it*$_x$ takes some course.
Every *thing*$_x$ is such that if *it*$_x$ is a student, then some *thing*$_y$ is such that *it*$_y$ is a course and *it*$_x$ takes *it*$_y$
$\forall x(Px \to \exists y(Qy \land Rxy))$

Consider now the sentence:

Every student takes every course

We now paraphrase the sentence in stages:

```
*Translation key*\
$P$ \_ : \_ is a student\
$Q$ \_ : \_ is a course\
$R$ \_ \_ : \_ takes \_
```

Every *thing*$_x$ is such that if *it*$_x$ is a student, then *it*$_x$ takes every course.
Every *thing*$_x$ is such that if *it*$_x$ is a student, then every *thing*$_y$ is such that if *it*$_y$ is a course, then *it*$_x$ takes *it*$_y$.
$\forall x(Px \to \forall y(Qy \to Rxy))$.

For another example, consider the sentence:

Some course is taken by every student.

We now paraphrase the sentence in stages:

```
*Translation key*\
$P$ \_ : \_ is a student\
$Q$ \_ : \_ is a course\
$R$ \_ \_ : \_ takes \_
```

Some *thing*$_x$ is such that *it*$_x$ is a course and every student takes *it*$_x$.
Some *thing*$_x$ is such that *it*$_x$ is a course and every *thing*$_y$ is such that if *it*$_y$ is a student, then *it*$_x$ takes *it*$_y$
$\exists x(Qx \wedge \forall y(Py \to Rxy))$

Issues with Translation

Translation into quantificational logic is subject to its own special difficulties. We distinguish at least two families of problems.

Special Quantifier Phrases Some quantificational structures pose special problems.

A A whale lives underwater

Quantifier phrases of the form 'a P' may on occasion be translated with the help of a universal quantifier and they may on other occasions be translated with the help of an existential quantifier. Consider the contrast between typical utterances of the sentences below:

A whale lives underwater.
A student aced the midterm.

The first sentence is generally used to express a universal claim:

```
*Translation key*\
$P$ \_ : \_ is a whale\
$Q$ \_ : \_ lives underwater
```

A whale lives underwater.
Every *thing* is such that if *it* is a whale, then *it* lives underwater.
$\forall x(Px \to Qx)$

Translation key:

P _ : _ whale
Q _ : _ lives underwater

On the other hand, the second sentence is generally used to express an existential claim:

```
*Translation key*\
$P$ \_ : \_ is a student\
$R$ \_ \_ : \_ aces \_ \
$a$: The Midterm
```

A student aced the midterm.
Some *thing* is such that *it* is a student and *it* aced the midterm.
$\exists x(Px \wedge Rxa)$.

Any Any philosopher thinks.

The quantifier phrase 'any' may be translated with the help of a universal or an existential quantifier in different cases.

Compare the sentences:

Any philosopher thinks.
If any philosopher thinks, then Hegel does.

The first sentence invites the use of a universal quantifier.

```
*Translation key*\
$P$ \_ : \_ is a philosophert\
$Q$ \_ : \_ thinks
```

- Any philosopher thinks.
- Every *thing* is such that if *it* is a philosopher, then *it* thinks.
- $\forall x(Px \to Qx)$

On the other hand, the second sentence suggests an existential quantifier:

```
*Translation key*\
$P$ \_ : \_ is a philosophert\
```

```
$Q$ \_ : \_ thinks\
$a$: Hegel
```

If any philosopher thinks, then Hegel does.
If some *thing* is such that *it* is a philosopher and *it* thinks, then Hegel thinks.
$\exists x(Px \wedge Qx) \rightarrow Qa$

Only Only seniors are eligible for the prize

How should we paraphrase this sentence? Here is a suggestion:

```
*Translation key*\
$P$ \_ : \_ is a senior\
$Q$ \_ : \_ is eligible for the prize
```

All eligible for the prize are students
Every *thing* is such that if *it* is eligible for the prize, then *it* is a senior
$\forall x(Qx \rightarrow Px)$

The interaction between quantification and negation requires some care as well:

Consider the sentence:

No student is eligible for the prize.

There is a choice between two equivalent interpretations. Consider, first, the sentence:

```
*Translation key*\
$P$ \_ : \_ is a senior\
$Q$ \_ : \_ is eligible for the prize
```

Every student is not eligible for the prize.
Every *thing* is such that if *it* is a student, then *it* is *not* eligible for the prize.
$\forall x(Px \rightarrow \neg Qx)$

For an alternative interpretation of the original sentence, consider:

It is not the case that some student is eligible for the prize.
It is not the case that some *thing* is such that *it* is a student and *it* is eligible for the prize.
$\neg\exists x(Px \wedge Qx)$

Structural Ambiguity A sentence may be structurally ambiguous.

Consider the sentence:

Everything has a cause

There is a choice between two non-equivalent interpretations:

```
*Translation key*\
$R$ \_ : \_ is a cause of \_
```

Every *thing* is such that it_1 has some cause.
Every *thing$_x$ is such that some thing$_y$ is such that it$_y$ causes it$_x$.
$\forall x \exists y Ryx$

The alternative interpretation proceeds as follows:

```
*Translation key*\
$R$ \_ : \_ is a cause of \_
```

Some thing$_x$ is such that it$_x$ causes everything.
Some thing$_x$ is such that every thing$_y$ is such that it$_x$ causes t_y.
$\exists x \forall y Rxy$

The distinction is specially important when it comes to the evaluation of an argument such as the following:

1. Everything has a cause.
2. If everything has a cause, then God is the cause of everything.
3. God is the cause of everything.

The crucial observation is that premise two is plausible on one but not the other interpretation of the first premise.

Consider the sentence:

Every student registers for some course

There is a choice between two *non-equivalent* interpretations:

```
*Translation key*\
$P$ \_ : \_ is a student\
$Q$ \_ : \_ is a course\
$R$ \_  \_ : \_ registers for \_
```

Every $thing_x$ is such that if it_x is a student, then it_x registers for some course.
Every $thing_x$ is such that if it_x is a student, then some thing$_y$ is such that it_y is a course and it_x registers for it_y.
$\forall x(Px \to \exists y(Qy \wedge Rxy))$

The alternative interpretation is this:

```
*Translation key*\
$P$ \_ : \_ is a student\
$Q$ \_ : \_ is a course\
$R$ \_ \_ : \_  registers for \_
```

Some thing$_x$ is such that it$_x$ is a course and every student registers for *it*$_x$.
Some thing$_x$ is such that it$_x$ is a course and every thing$_y$ is such that if *it*$_y$ is a student, then *it*$_y$ registers for *it*$_x$.
$\exists x(Q(x) \wedge \forall y(Py \to Ryx))$

We are in a position to combine the skills we have gained in order to determine whether an argumnet formulated in English exemplifies a valid quantificational form.
Example 9.1. Consider the argument:

1. All physicists admire Newton.
2. Newton admires some mathematicians.
3. So, some physicists admire some mathematicians.

We first translate the argument into the language of quantificational logic:

```
*Translation key*\
$P$ \_ : \_ is a physicist\
$Q$ \_ : \_ is a mathematician\
$R$ \_ \_ : \_ admires \_
```

1. $\forall x(P(x) \to R(x, a))$
2. $\exists x(Q(x) \wedge R(a, x))$
3. $\exists x(P(x) \wedge \exists y(Q(y) \wedge R(x, y)))$

The next step is to search for a model in which the premises are true and the conclusion false:

```
$P$ \_ : \_  is enclosed by a circle \
$Q$ \_ : \_ is enclosed by a diamond \
$R$ \_ \_ : \_ points to \_\
$a : p_1$
```

Example 9.2. Consider the argument:

1. Some LA landmark is moore beautiful than some NY landmark.
2. The Griffith Observatory is an LA landmark and Times Square is a NY landmark.
3. So, the Griffith Observatory is more beautiful than Time Square.

We first translate the argument into the language of quantificational logic:

```
*Translation key*\
$P$ \_ : \_ is an LA landmark\
$Q$ \_ : \_ is a NY landmark\
$R$ \_ \_ : \_ is more beautiful than \_
```

1. $\exists x(Px \wedge \exists y(Qy \wedge Rxy))$
2. $Pa \wedge Qb$
3. Rab

The next step is to search for a model in which the premises are true and the conclusion false:

```
$P$ \_ : \_ enclosed by a circle \
$Q$ \_ : \_ is enclosed by a diamond \
$R$ \_ \_ : \_ points to \_ \
$a : p_1$\
$b: p_2$
```

Exercises

1. Translate the following sentences into the language of quantificational logic.

   a. Some Trojans fear Achilles, but Achilles fears every Trojan.

   b. All Trojans are courageous.

   c. Only Trojans come from Troy.

   d. No Greek comes from Troy.

   e. Achilles fears Paris, but Paris fears no one.

   f. Some Greeks fears some Trojans, and some Trojans fear some Greeks.

2. Assess the following arguments for quantificational validity.

   a. All poisonous frogs are amphibians. Kermit is a posisonous amphibian. Therefore, Kermit is a frog.

   b. Aristotle admired anyone who taught for free. Socrates taught for free, but no sophist did ever teach for free. So, Aristotle did not admire a sophist.

   c. No weekday comes after Saturday. Monday is a weekday, but Sunday is not. Therefore, Monday does not come after Sunday.

   d. Some cube is green all over and some cube is red all over. No cube is both green all over and red all over. Some sphere is green all over. So, some sphere is not red all over.

   e. No Trojan fears Achilles, but Achilles fears some Trojans. Paris is a Trojan. So, Achilles fears Paris but Paris does not fear Achilles.

f. Some stadium is close to University Park. The Coliseum is a stadium. Therefore, the Coliseum is close to University Park.

g. Los Angeles is connected by direct flight to London. London is connected by direct flight to every major European city. Barcelona is a major European city. Therefore, Los Angeles is connected by direct flight to Barcelona.

h. All Greeks feared Hector or Paris. Paris feared no Greek who feared Hector. Therefore, some Greeks feared Paris.

10

Natural Deduction

We have effective means to determine, in a finite number of steps, whether an argument formulated in propositional logic is valid. We may, for example, devise a complete truth table for the argument and look for an assignment corresponding to one of the rows of the truth table on which the premises are true and the conclusion is false. If no such assignment is found, then the argument is valid; otherwise, the argument is invalid. The procedure may be overly complex in certain cases, especially when the argument in question happens to involve a high number of propositional variables, but it will nevertheless produce a verdict in a *finite* number of steps.

Matters are more complicated for quantificational logic. For we are not of course in a position to survey *every* model for the relevant fragment of the language in order to determine wether or not there is one on which the premises true and the conclusion false. There is an infinite number of them, and there is no effective procedure we can use to survey them all in a finite number of steps. To be sure, if the argument is invalid, we should be able to find a model in which the premises are true and the conclusion is false, but we may not be sure at times whether a temporary failure to find such a model may be due to a lack of imagination rather than to the invalidity of the argument.

Instead, we will develop a natural deduction system for quantificational logic, one which adapts and extends the framework we developed for propositional logic. Once the natural deduction system is in place, we will establish the validity of an argument in the language of quantificational logic by means of a proof of the conclusion from the premises of the argument.

Since quantificational logic includes the usual propositional connectives, we will just import and adapt the introduction and elimination rules we devised for them to the new framework. That is enough to enable us to establish the validity of *some* arguments in the language of quantificational logic.

Example 10.1. $\forall x Px \to \exists x Px, \neg \exists x Px \vdash \neg \forall x Px$.

Notice that it suffices in this case to deploy natural deduction rules for negation and the conditional, which we adapt from the natural deduction system for propositional logic.

On the other hand, we are not yet in a position to provide derivations that exploit the behavior of the quantifiers. We require additional natural deduction rules for that purpose.

Universal Quantification

We will introduce introduction and elimination rules for the universal and the existential quantifiers. But in order to even be able to state the rules, we must explain what we mean by an *instance* of a quantified formula of the form $\forall x \varphi$ or $\exists x \varphi$.

Instance If a is a constant, we let $\varphi(a/x)$ be the formula that results from φ when we substitute a for every *free* occurrence of the variable x in φ. The formula $\varphi(a/x)$ is an *instance* of the quantified formulas $\forall x \varphi$ and $\exists x \varphi$, respectively.

Example 10.2. Some examples:

- Pc is an instance of the universal quantification $\forall x Px$.
- Ray is an instance of the universal quantification $\forall x Rxy$.
- $Rzbb$ is an instance of the universal quantification $\forall y Rzyy$.
- $Pb \to \exists x Qx$ is an instance of the existential quantification $\exists x(Px \to \exists x Qx)$.

We start now with the elimination rule for the universal quantifier.

```
<img src="images/ue.png">\
  Universal Elimination
```

Universal Elimination You may write an instance $\varphi(a/x)$ of a universally quantified formula $\forall x \varphi$ if the latter is available on a prior line of the derivation.

The rationale for universal elimination is that we are entitled to write that an individual a satisfies a given condition if we are given that *all* individuals do.

Let us look at the universal elimination rule in action.

Example 10.3. $\forall x \forall y(Px \to Qy), Pa, \vdash Qb$.

Example 10.4. $\forall x \forall y Rxy \vdash Raa$.

The introduction rule for the universal quantifier requires more care. In order to be able to write $\forall x\varphi(x)$ by itself on a line, we should convince ourselves we have the means to establish some *arbitrary* instance $\varphi(a/x)$ where the constant a does not occur in an undischarged assumption.

```
<img src="images/ui.png">\
  Universal Introduction
```

Universal Introduction You may write a universally quantified formula $\forall x\varphi$ if some instance $\varphi(a/x)$ is available on a prior line and a occurs neither in φ nor in an undischarged assumption.

Let us look at some applications of the rule.

Example 10.5. $\forall x(Px \to Qx), \forall x(Qx \to Rx) \vdash \forall x(Px \to Rx)$.

Example 10.6. $\forall x(Px \to \neg Px) \vdash \forall x \neg Px)$

Existential Quantification

We now introduce introduction and elimination rules for the existential quantifier.

```
<img src="images/ei.png">\
  Existential Introduction
```

Existential Introduction You may write an existentially quantified formula $\exists x\varphi$ provided an instance $\varphi(a/v)$ is available on a prior line of the derivation

The rationale for existential introduction is that we are entitled to write that something satisfies a condition if we are given that an individual a does.

Here is the rule of existential introduction in action.

Example 10.7. $Rab \vdash \exists x\exists y Rxy$

Example 10.8. $\neg\exists x(Px \wedge Qx) \vdash \neg\forall x(Px \wedge Qx)$

The elimination rule for the existential quantifier is more subtle. Suppose we want to reason from an existential generalization of the form $\exists x\varphi(x)$ to some formula ψ. The elimination rule for the existential tells us that the step is justified if we have the means to establish a conditional $\varphi(a/v) \to \psi$ where a is a constant that occurs neither in ψ nor $\exists x\varphi$ nor in an earlier open assumption.

```
<img src="images/ee.png">\
  Existential Elimination
```

Existential Elimination You may write ψ if the existentially quantified formula $\exists x\varphi$ and the conditional $\varphi(a/x) \to \psi$ are available at prior lines provided that a occurs neither in neither in ψ nor in $\exists x\varphi$ nor in an undischarged assumption.

Example 10.9. $\exists x(Px \land \neg Qx) \vdash \neg\forall x(Px \to Qx)$

Example 10.10. $\forall x(Px \to Qx), \exists yPy \vdash \exists zQz$

Common Mistakes and Strategies

Much of the advice for the construction of proofs in propositional logic carries over to quantificational logic. In order to prove a universal or existential generalization from a set of premises, it helps to work backwards and with a view to being in a position to use the relevant introduction rule. And to reason forward from a universal or existential generalization, we should use the relevant elimination rules.

One important observation is that the elimination rule for the universal quantifier should not be applied to formulas that are *not* universally quantified formulas. The rule may be applied, for example, to the universal quantification $\forall x(Px \to Qx)$ to obtain an instance such as $Pa \to Qa$. It would, however, be a mistake to attempt to apply the rule to the formula $\forall xPx \to \forall xQx$, which, unlike the first, is just a conditional. The key is to be aware of what is the main connective in the relevant formula and to make sure to apply the rule that corresponds to it.

One more potential pitfall is the failure to apply the rules for quantification once at a time. The elimination rule for the universal quantifier does *not* allow an immediate transition from a formula such as $\forall x\forall yRxy$ to Rab. We should instead distinguish two steps: one from the original formula to $\forall yRxy$ and a further one from this formula to the instance Rab.

Let us now look at some further examples.

Example 10.11. $\exists x\forall yRxy, \forall x\forall y(Rxy \to Ryx) \vdash \exists x\forall yRyx$

Example 10.12. $\forall x(Px \to (Qxa \to Rxb)) \vdash \forall x(Px \to Qxa) \to \forall x(Px \to Rxb)$

Exercises

1. Provide a natural deduction proof in order to justify each of the claims below:

   a. $\exists xPx \to \exists xQx, \forall xPx \vdash \exists xQx$

   b. $\exists xPx \land \exists xQx, \neg\exists x(Px \land Qx), \forall x(Rx \to Px) \vdash \exists x\neg Rx$

   c. $\forall x(Px \to \neg Qx), \forall x(Rx \to Qx) \vdash \forall x(Rx \to \neg Px)$

   d. $\forall x(Px \to (\exists yRxy \to Qx)), \neg\exists yQy \vdash \forall y(Py \to \forall z\neg Ryz)$

e. $\forall x(Px \rightarrow \forall y(Rxy \rightarrow Py)), \exists x(Qx \wedge Px) \vdash \exists x(Qx \wedge \forall y(Rxy \rightarrow Py))$

11

Identity

We begin with a distinction between two important binary relations: numerical and qualitative identity. The relation of numerical identity is one an object bears to itself and to nothing else, and it is generally expressed by means of the predicate 'is' or 'is identical to'. Unfortunately, that predicate is sometimes used to express a different relation, one that obtains between two objects if they are similar in every respect that may be relevant for the purposes of a conversation. We use the label 'numerical identity' for the first relation and the label 'qualitative identity' for the second.

Qualitative Identity Two objects are qualitatively identical when they share all of the same qualities.

If Alex and Beth bought two computers of the same make and with the same specifications and even the same color, we may be inclined to say:

- Alex and Beth bought (qualitative) identical computers.
- The computer Alex bought is identical to the computer Beth bought.
- Alex and Beth bought the same computer.

Numerical Identity We speak of numerical identity when two objects are one and the same.

If Alex and Beth do not own different computers but rather happen to share one and the same machine, we may be inclined to say:

- The computer Alex uses is (numerically) identical to the computer Beth uses.
- Alex and Beth use (one and) the same computer.
- Alex and Beth share (one and) the same computer.

Unfortunately, a variety of sentences are ambiguous between two interpretations of the relevant predicates:

- I have always used the same computer.
- The computer I bought last year is the same you bought now.

On the interpretation of the predicate 'the same' in terms of numerical identity, one and the same computer has been used throughout the years and one and the same computer had been bought by me one year ago and has been bought by you now. On the alternative interpretation in terms of qualitative identity, I have always used computers of the same make and specifications even if more than one, and the computer you bought now is of the same make and specifications as the one I had bought one year ago.

Syntax

Quantificational logic with identity expands the vocabulary of quantificational logic with a predicate $=$ for numerical identity, which is interpreted to mean: *one and the same object*. All formulas of quantificational logic are formulas of quantificational logic with identity, but the definition of atomic formula of the latter is now expanded:

Atomic Formula All atomic formulas of quantificational logic remain atomic formulas of quantificational logic with identity. Furthermore, if τ_1 and τ_2 are two constants or variables, then

$$\tau_1 = \tau_2$$

is an *atomic formula* of quantificational logic and identity.

Example 11.1. Each of the expressions below are atomic formulas of quantificational logic with identity:

$$\begin{gathered} x = y \\ a = x \\ a = b \\ \ldots \end{gathered}$$

Formula We now define what is for an expression to be a formula of quantificational logic with identity.

1. All atomic formulas are *formulas*.

2. If φ and ψ are *formulas*, then each of

$$\neg\varphi, (\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \to \psi)$$

   is a *formula*.

3. If φ is a formula and x is a variable, then each of $\forall x\varphi$ and $\exists x\varphi$ is a formula.

4. Nothing else is a *formula*.

All other definitions carry over without incident to the language of quantificational logic with identity.

Semantics

The semantics for quantificational logic with identity requires a minimal modification of the semantics for quantificational logic. We retain the definition of a model, an assignment for a model, and that of the denotation of a singular term relative to an assignment. We modify the definition of truth in a model relative to an assignment in order to accommodate atomic formulas of quantificational logic with identity.

Atomic Formulas

$\tau_1 = \tau_2$ is true in a model M under an assignment α if the denotation of τ_1 relative to α is one and the same object as the denotation of τ_2 relative to α.

Example 11.2.

```
$D = \{1, 2\}$\
$I(P) = \{1, 2\}$\
$I(Q) = \{2\}$\
$I(R) = \{\langle 1, 2\rangle\}$\
$I(a) = 1$\
$I(b) = 2$
```

Let α be an assignment for the model M:

x	y	z	x_1	y_1	z_1	...
1	2	2	2	2	1	...

Then:

- $b = y$ is true in M relative to α because $I(b)$ is one and the same object as $\alpha(y)$, namely, 2.
- $x = y$ is *not* true in M relative to α because $\alpha(x)$ is *not* the same object as $\alpha(y)$.
- $a = b$ is *not* true in M relative to α because $I(a)$ is *not* the same object as $I(b)$.

The rest of the definition of truth in a model relative to an assignment and truth in a model carries over to quantificational logic with identity without incident. So do the definitions of logical truth, equivalence, consistency, and validity.

```
<img src="images/ex 9.4.png" style="display:block; margin: 0 auto;">\

$P: \text{enclosed by a circle}$ \
$a : p_1$ \
$b: p_2$
```

Example 11.3. The formula $(Pa \to Pb) \to \forall x(x = a \to x = b))$ is *not* a logical truth.

The model depicted by the diagram verifies the antecedent of the conditional but not the consequent.

- The formula $Pa \to Pb$ is true in the model because p_1 and p_2 are each enclosed by a circle.

- The formula $\forall x(x = a \to x = b)$ is not true in the model because $x = a \to x = b$ is not true no matter what we assign to x. That is because a and b denote different objects in the model.

```
<img src="images/ex 9.6.png" style="display:block; margin: 0 auto;">\

$P: \text{enclosed by a circle}$ \
$a : p_1$ \
$b: p_1$
```

Example 11.4. Consider the following argument:

1. $\forall x(x = a \to Px)$
2. $a = b$
3. $\forall x(Px \to x = b)$

The model depicted by the diagram verifies the premises but falsifies the conclusion.

- The formula $\forall x(x = a \to Px)$ is true in the model because $(x = a \to P^1 x)$ is true no matter what we assign to x: if we assign p_1 to x, then both antecedent and consequent come out true, but if we assign p_2 to x, the antecedent is false making the conditional true.

- The formula $a = b$ is true in the model because the denotation of a in the model is one and the same object as the denotation of b in the model.

- The formula $\forall x(Px \to x = b)$ is false in the model because $(Px \to x = a)$ is not true when we assign p_2 to x: p_2 is enclosed by a circle, but it is not one and the same object as the denotation of a. We conclude that the argument is invalid.

```
<img src="images/ex 9.7.png" style="display:block; margin: 0 auto;">\
```

```
$R: \text{points to}$ \
$a : p_1$ \
$b: p_2$ \
$c: p_3$ \
$d: p_1$
```

Example 11.5. Consider the following argument:

1. $Rab \wedge \neg Rac$
2. $\exists x Rcx \wedge \neg Rad$
3. $\neg a = d$

This model depicted by the diagram verifies the premises but not the conclusion.

- The formula $Rab \wedge \neg Rac$ is true in thhe model because the denotation of a in the model, namely, p_1, points to the denotation of b in the model, namely, p_2, but p_1, does *not* point to the denotation of c in the model, namely, p_3.

- The formula $\exists x Rcx \wedge \neg Rad$ is true in the model because the denotation of c in the model, namely, p_3, points to something but the denotation of a in the model, namely, p_1, does not point to the denotation of d in the model, namely p_1.

- The formula $\neg a = d$ is false in the model because the denotation of a in the model is one and the same object as the denotation of d in the model, namely, p_1.

We conclude that the argument is invalid.

Natural Deduction

We now expand the natural deduction system for quantificational logic with introduction and elimination rules for identity.

There is a simple introduction rule for identity:

```
<img src="images/idi.png">\
  Identity Introduction
```

Identity Introduction You may write an identity $\tau = \tau$ by itself on a line.

The rationale for this rule is that since as a matter of logic, all objects are self-identical, whatevery τ may be, it will likewise identical to itself.
Example 11.6. $\forall x(x = x \rightarrow x = b) \vdash \forall x\ x = b$.

```
<img src="images/ide.png">\
  Identity Elimination
```

Identity Elimination You may substitute every occurrence of τ_2 for τ_1 in a formula φ when both that formula and an identity $\tau_1 = \tau_2$ occur at prior available lines of a proof

The rationale for the elimination rule is that given $\tau_1 = \tau_2$, τ_1 is one the same object as τ_2, which means that whatever is true of one should remain true of the other.

Example 11.7. $\forall x(\neg Rbx \to (Px \to x = b)), Rab \wedge \neg Raa, Rba \to \neg Rab \vdash \neg Pa$

Uses of Identity

We will use identity to translate overt identity statements such as 'Mark Twain is Samuel Clemens', which becomes $a = b$ relative to a translation key on which a translates 'Mark Twain' and b translates 'Samuel Clemens'. That use of the word 'is' to express numerical identity is to be contrasted with uses of the word on which it expresses a predication, e.g., 'Mark Twain is a writer' which would be translated P^1a relative to a translation key on which a translates 'Mark Twain' and P translates 'is a writer'.

We now look at less overt uses of identity in natural language.

Only Only you arrived on time

```
*Translation key*\
$P$: arrived on time\
$a$: you
```

Every *thing* is such that *it* arrived on time only if *it* is identical to you.

$\forall x(Px \to x = a)$

Else You arrived on time, but someone else did as well.

```
*Translation key*\
$P$: arrived on time\
$a$: you
```

You arrived on time, and some *thing* is such that *it* is not identical to you and *it* arrived on time.

$Pa \wedge \exists x(\neg x = a \wedge Px)$

Number Identity enables us to translate a variety of statements of number.

Some statements of number say that there is at least a certain number of objects of a certain type:

- There are *at least* two books.
- There are *at least* three books.

Others tell us that there are at most a certain number of such objects:

- There is *at most* one book.
- There are *at most* two books.

There is of course no reason to use identity to translate 'there is at least one book' as $\exists xPx$ relative to a translation key on which P translates 'book'. On the other hand, neither $\exists x\exists y(Px \wedge Py)$ nor $\exists xPx \wedge \exists yPy$ are adequate translations of the statement 'there are at least two books' relative to that translation key. For they merely state that something is a book and something is a book, which is compatible with one and the same witness for each existential generalization. We do better with the help of identity, since we are able to offer the following translations for sentences in the first group:

- $\exists x\exists y(Px \wedge Py \wedge \neg x = y)$
- $\exists x\exists y(Px \wedge Py \wedge Pz \wedge \neg x = y \wedge \neg x = z \wedge \neg y = z)$

It is not difficult to generalize to other numbers, e.g., 'there are at least four books', etc. We are similarly able to offer the following translations for the sentences in the second group:

- $\forall x\forall y(Px \wedge Py \to x = y)$
- $\forall x\forall y\forall z(Px \wedge Py \wedge Pz \to x = y \vee x = z \vee y = z)$

The generalization to other numbers is similarly straightforward.

We are now in a position to translate numerical statements whereby we state the existence of an exact number of objects of a certain type:

- There is *exactly* one book.
- There are *exactly* two books.

The crucial observation at this point is that each of these statements is equivalent to a conjunction of two statements we already know how to translate:

- There is at least one book and there is at most one book.
- There are at least two books and there are at most two books.

The translation of each of these conjunctions becomes:

- $\exists xPx \wedge \forall x\forall y(Px \wedge Py \to x = y)$
- $\exists x\exists y(Px \wedge Py \wedge \neg x = y) \wedge \forall x\forall y\forall z(Px \wedge Py \wedge Pz \to x = y \vee x = z \vee y = z)$

But each of these formulas is equivalent to a more concise version:

- $\exists x(Px \wedge \forall y(Py \to x = y))$
- $\exists x \exists y(Px \wedge Py \wedge \neg x = y \wedge \forall z(Pz \to x = z \vee y = z))$

Definite Descriptions The author of *Huckleberry Finn* is American

We now focus on phrases such as 'the author of *Huckleberry Finn*' or 'instructor', which combine the expression 'the' with a predicate. In quantificational logic, we treated all such expressions as designators, whose primary semantic function is to designate an object. But Bertrand Russell famously suggested that they should instead be treated as quantifier phrases. He offered the following style of paraphrase for sentences involving definite descriptions:

- The instructor arrived on time.
- Exactly one individual is an instructor and she arrived on time.
- The author of *Huckleberry Finn* is American
- Exactly one individual is author of *Huckleberry Finn* and he is American.

Given Russell's paraphrase method, the first sentence translates as:

- $\exists x(Px \wedge \forall y(Py \to x = y) \wedge Qx)$,

relative to a translation key on which P^1 translates 'instructor' and Q^1 translates 'arrived on time'. Similarly, the second sentence now translates:

- $\exists x(Rxa \wedge \forall y(Rya \to x = y) \wedge Px)$,

relative to a translation key on which a tranlates '*Huckleberry Finn*', R translates 'authors' and P translates 'American'.

Exercises

1. For each set of formulas below, find a model we can use to justify the fact that the set is consistent. Please justify your answers.

   a. $\{\forall x(Px \leftrightarrow Qx), Pa, Qb, \neg a = b\}$

   b. $\{\forall x(Rax \to Rbx, \neg Raa, Rba\}$

2. For each argument below, find a model we can use to justify the fact that the argument is invalid. Please justify your answers.

   a. $\forall x(Px \to Qx), \neg Qa \vdash a = b \to Rab$

   b. $\exists x Px, \exists x Qx \vdash \exists x \exists y \neg x = y$

3. Provide a natural deduction proof to justify each of the claims below:

   a. $\forall x(Px \rightarrow Qx), \neg Qa, Pb \vdash a = b \rightarrow Rab$

   b. $\exists x \forall y(Px \leftrightarrow y = x), Pa \vdash \forall x(Px \rightarrow x = a)$

   c. $\exists x Px, \exists x Qx, \neg \exists x(Px \wedge Qx) \vdash \exists x \exists y \neg x = y$

Part IV

Appendix

12

Solutions to Selected Exercises

1. Reason and Argument

1. Identify the premises and conclusion of each of the following arguments.

   a. You did not turn the ignition. The car starts when you turn the ignition and the tank is full. But the tank is full and the car did not start.

   b. If the mind is the same as the brain, then whatever is true of one is true of the other. The brain is a physically extended object but the mind is not. So, the mind is not the same as the brain. **Solution**:

      1. If the mind is the same as the brain, then whatever is true of one is true of the other.
      2. The brain is a physically extended object but the mind is not.
      3. The mind is not the same as the brain.

   c. The deal will fall through unless Smith is part of the negotiation. So, the deal will not be completed. For Smith just phoned to let us know she cannot join the negotiation. **Solution**:

      1. The deal will fall through unless Smith is part of the negotiation.
      2. Smith just phoned to let us know she cannot join the negotiation.
      3. The deal will not be completed.

   d. You will be admitted into the club just in case you apply for club membership and you are not perceived to be eager to join. Applying for club membership is regarded as a sign of eagerness. That is, if you apply for club membership, you will be perceived to be eager to join. So, you will not be admitted into the club. **Solution**:

1. You will be admitted into the club just in case you apply for club membership and you are not perceived to be eager to join.
2. If you apply for club membership, you will be perceived to be eager to join.
3. You will not be admitted into the club.

2. Which of the following arguments are valid?

   a. It today is Tuesday, then today is not Wednesday. Today is Tuesday. So, today is not Wednesday. **Solution**: This is a valid argument because it is an instance of a valid argument pattern: If p, then not-q p Not-q and none of its instances has true premises and a false conclusion.

   b. Today is Tuesday or Thursday. So, today is Tuesday **Solution**: This is *not* a valid argument because it is not an instance of a valid argument pattern. It is, for example, an instance of the pattern: p or q p To appreciate its invalidity, notice that there are instances with true premises and a false conclusion, e.g. Snow is black or snow is white (*true*) Snow is black (*false*).

   c. Today is Tuesday or Thursday. Today is not Thursday. So, today is Tuesday

   d. I'm late on Tuesdays. I'm late again today. So, today is Tuesday

3. Justify the invalidity of each of the argument patterns below by providing an instance with true premises and a false conclusion.

   a. If p, then q or r.
   Not q.
   Therefore, r.

   b. p unless q.
   q.
   p. **Solution**: This is *not* a valid argument pattern because some of its instances have true premises and a false conclusion, e.g.,1. The commute to campus will take over an hour unless I find a way to live nearby campus. (*true*)2. I will find a way to live nearby campus (*true*)3.The commute to campus will take over an hour (*false*).

   c. p if, and only if, both q and r.
   r.
   q.

4. True or false? Justify your answers.

   a. There are valid arguments with false premises and a true conclusion. **Solution**: This is *true*. The following is a valid argument with false premises and a true conclusion:1. Snow is white and snow is black (*false*)2. Snow is white (*true*).It is a valid argument because it is an instance of a valid argument pattern:p and $q.p$.

b. There are invalid arguments with true premises and a true conclusion.

c. There are valid arguments with a false conclusion.

d. There are sound arguments with a false conclusion.**Solution**: This is *false*: a sound argument is a valid argument with true premises. But if a valid argument has true premises, then it must have a true conclusion. It follows that a sound argument must have a true conclusion.

e. If an argument is valid, then at least one of its premises is true.**Solution**: This is *false*. The argument we deployed in response to part a has *no* true premises at all.

2. Propositional Logic

1. Add quotation marks to the following sentences in order to obtain a true sentence of Enlglish. If there is more than one solution, make sure to explain why.

   a. Los Angeles names a city in Southern California.

   b. Los is part of Los Angeles. **Solution**: 'Los' is part of 'Los Angeles'

   c. Snow is white is a grammatical sentence of English. **Solution**: 'Snow is white' is a grammatical sentence of English.

   d. What is monosyllabic but whatever is polysyllabic. **Solution**: 'What' is monosyllabic but 'whatever' is polysyllabic..

2. Determine whether each of the following expressions is a sentence of propositional logic.

   a. $\neg\neg A$

   Solution:

```
<img src="images/exex 2.2a.png">

No construction tree is available.
```

```
<span style="color:green;">This  is *not* a sentence of propositional logic.<br/>
By rule 2, $\neg \neg A$ is a sentence if $\neg A$ is a sentence.<br/>
By rule 2, $\neg A$ is a sentence  if $A$ is a sentence.<br/>
```

```
    By rule 3, since $A$ is *not* a propositional variable, $A$ is not a sentence.<br/> So, it follow

#. $(p \to (\neg q))$

#. $(p)$

#. $(q \to (\neg  \neg  p \vee r))$

  <br/>**Solution**:<br/>
```

```
<img src="images/exex 2.2d.png">

A construction tree for the formula.
```

```
    <span style="color:green;">This *is* a sentence of propositional logic.<br/>
    By rule 2, $(q \to (\neg  \neg  p \vee r))$ is a sentence if both $q$ and $(\neg \neg p \vee r)$
    By rule 1, $q$ is a sentence.<br/>
    By rule 2, $(\neg \neg  p  \vee r)$ is a sentence if both $\neg \neg p$ and $r$ are sentences.<br
    By rule 2, $\neg  \neg p$ is a sentence if $\neg p$ is a sentence.<br/>
    By rule 2, $\neg  p$ is a sentence  if $p$ is a sentence.<br/>
    By rule 1, both $p$ and $r$ are sentences. <br/> So, we conclude that $(q \to (\neg  \neg  p \vee
```

1. Determine whether each of the following expressions is an *abbreviation* for a sentence of propositional logic.

   a. $p \to (q \wedge r)$**Solution**: By convention 3, this is an abbreviation for $(p \to (q \wedge r))$.

   b. $p \to q \to r$

   c. $p \vee (q \wedge r)$

   d. $p \to (q \wedge r \to q)$

2. Use the truth table method to determine whether each of the following formulas is a *tautology*, a *contradiction*, or neither a tautology or a contradiction.

   a. $((p \vee q) \to \neg(p \to q))$**Solution**: This sentence is *neither* a tautology *nor* a contradiction because it is true under some assignments, e.g., rows 2 and 4, and false under others, e.g., rows 1 and 3.

p	q	$(p \vee q)$	$\neg(p \to q)$	$((p \vee q) \to \neg(p \to q))$
T	T	T	F	F
T	F	T	T	T
F	T	T	F	F
F	F	F	F	T

b. $p \to (q \to p)$
c. $q \to (q \to p)$

3. Use the truth table method to determine whether the following pairs of sentences are logically equivalent.

a. $p \to q$, $q \wedge \neg(p \vee q)$

b. $p \to p$, $q \to q$ **Solution**: These sentence are logically equivalent because they have the same truth value under every assignment.

p	q	$p \to p$	$q \to q$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	T	T

4. Use the truth table method to determine whether the following sets of sentences are consistent.

a. $\{p, q, \neg(p \wedge q)\}$

b. $\{p \vee q, q \to r, \neg(p \wedge r)\}$ **Solution**: This is a consistent set of sentences, since every sentence in the set is true under some assignments, e.g., rows 4 and 5.

p	q	r	$p \vee q$	$q \to r$	$\neg(p \wedge r)$
T	T	T	T	T	F
T	T	F	T	F	T
T	F	T	T	T	F
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	T	F	T
F	F	T	F	T	T
F	F	F	F	T	T

5. Use the truth table method to determine whether the following arguments are valid.

a. $p \to q$
$q \to r$
r

b. $q \vee (p \wedge r)$
$\neg q$
r **Solution**: This argument is *valid* since there is no assignment, that is, no row of the truth table, on which the premises come out

true and the conclusion false:

p	q	r	$q \vee (p \wedge r)$	$\neg q$	r
T	T	T	T	F	T
T	T	F	T	F	F
T	F	T	T	T	T
T	F	F	F	T	F
F	T	T	T	F	T
F	T	F	T	F	F
F	F	T	F	T	T
F	F	F	F	T	F

6. Use the search-for-counterexample method to determine whether the following arguments are valid.

   a. $p \rightarrow (q \vee r)$
   $r \rightarrow (p \rightarrow q)$
   p
   q **Solution**: The argument is *invalid*, since we are able to build an assignment under which the premises come out true and the conclusion false, e.g., one on which p is true but both q and r are false.

p	q	r	$p \rightarrow (q \vee r)$	$r \rightarrow (p \rightarrow q)$	p	q
T	F	F	T	T	T	F

   b. $(p \rightarrow q) \wedge (q \rightarrow p)$
   $\neg(p \wedge r)$
   $\neg(r \rightarrow s)$
   $s \rightarrow q$

3. Translation into Propositional Logic

1. Translate the following sentences into the language of propositional logic.

   a. If interest rates fall, then we will be able to borrow more money and inflation will rise. **Solution**:

```
*Translation key*\
$p$: Interest rates fall.\
$q$: We will be able to borrow more money.\
$r$: Inflation will rise.\
```

We proceed in three steps: If interest rates fall, then (we will be able to borrow more money and inflation will rise) Interest rates fall $\rightarrow$ (we will be able to borrow more money $\wedge$ inflation will rise) $p \rightarrow (q \wedge r)$

```
#.  Your computer will not start unless you turn it on and the battery is fully charged.

#.  The mayor's office made the decision, even though it did not sit well with its constituents.

#. You will carry an umbrella outside just in case it is raining or you are under the impression that
<br/>
<span style="color:green;">
```

```
*Translation key*\
$p$: Interest rates fall.\
$q$: We will be able to borrow more money.\
$r$: Inflation will rise.\
```

```
We proceed in three  steps:<br/>
<br/>
You carry an umbrella outside if, and only if, (it is raining or you are under the impression that it
<br/>
You carry an umbrella outside $\leftrightarrow$ (it is raining $\vee$ you are under the impression tha
<br/>
$p \leftrightarrow (q \vee r)$<br/><br/>
</span>

#. The deal will be completed if all parties agree to it and the deal is found to be in compliance wi
```

1. Assess the following arguments for propositional validity.

   a. The car will not start unless you turn the ignition and press the accelerator. You turned the ignition but the car failed to start. So, you did not press the accelerator. **Solution**: First, we rephrase the argument in premise conclusion form: 1. If it is not the case that you turn the ignition and press the accelerator, then the car will not start. 2. You turned the ignition and the car did not start. 3. You did not press the accelerator. Second, we translate the argument into the language of propositional logic:1. $\neg(p \wedge q) \rightarrow \neg r$2. $p \wedge \neg r$ 3. $\neg q$ Third, we use the search for counterexample method to check whether there is an assignment on which the premises come out true and the conclusion false:

p	q	r	$\neg(p \wedge q) \rightarrow \neg r$	$p \wedge \neg r$	$\neg q$
T	T	F	T	T	F

   Since we have identified one such assignment, we conclude that the argument is propositionally invalid.

   b. The mind is a simple substance only if it is not made of further parts. If the mind is material, then it is is made of further parts. So, if the mind is a simple substance, then it is not material.

c. My client is not the burglar. For the doors show no signs of forced entry. If the doors show no signs of forced entry, then if my client is the burglar, he must have climbed to the unlocked window on the second floor. However, my client is arthritic and could not climb to that unlocked window on the second floor.

d. If the British are coming, then they are coming by land or by sea. Paul Revere will light only one lamp, if they are coming by land. He will light two lamps, if they are coming by sea. Paul Revere will light only one lamp. So, the British are coming.

e. The insurance company will cover your expenses only if you are current with your monthly payments. But if you are current with your monthly payments, then you can afford to cover the expenses yourself. So, the insurance company will not cover your expenses unless you can afford to cover them yourself. **Solution**: First, we rephrase the argument in premise conclusion form: 1. The insurance company will cover your expenses only if you are current with your monthly payments. 2. If you are current with your monthly payments, then you can afford to cover the expenses yourself. 3. The insurance company will not cover your expenses unless you can afford to cover them yourself. Second, we translate the argument into the language of propositional logic: 1. $p \to q$2. $q \to r$ 3. $\neg r \to \neg p$ Third, we use the search for counterexample method to check whether there is an assignment on which the premises come out true and the conclusion false:

p	q	r	$p \to q$	$q \to r$	$\neg r \to \neg p$
T	T	F	T	!	F

Since no such assignment exists, we conclude that the argument is propositionally valid.

4. Natural Deduction for Propositional Logic

1. Provide a natural deduction proof in order to justify each of the claims below:

a. $p, \neg q \vdash p \to (\neg q \vee r)$ **Solution**:

b. $p \leftrightarrow (q \vee r), \neg q, \neg r \vdash \neg p$

c. $p \to \neg(q \wedge r), q, \neg r \to \neg q \vdash \neg p$ **Solution**:

d. $p \leftrightarrow q, q \vee r, r \to (p \vee q) \vdash q$

e. $(p \wedge q) \leftrightarrow (q \vee r), \neg r \to (\neg p \to q) \vdash p$ **Solution**:

5. Natural Language and Propositional Logic

1. Determine whether the following English sentences are tautologies, contradictions, or neither tautologies nor contradictions:

   a. Someone is a logician and not a logician. **Solution**: We translate the sentence into the language of propositional logic: p **Translation key**: p: Someone is a logician and not a logician. There is a simple an assignment on the sentence is false:

p	p
F	F

   We conclude it is not a logical truth.

   b. Some logicians are philosophers but some logicians are not philosophers.

   c. If all logicians are philosophers, then if some philosophers are logicians, then all logicians are philosophers.

   d. Someone is a logician and someone is not a logician. **Solution**: We translate the sentence into the language of propositional logic: $p \wedge q$ **Translation key**: p: Someone is a logician. q: Someone is not a logician. There is a simple an assignment on the sentence is false:

p	q	$p \wedge q$
F	F	F

   We conclude it is not a logical truth.

   e. All logicians are philosophers but no philosophers are logicians.

2. Determine whether the following arguments are propositionally valid:

   a. The deal will be completed only if Smith or Jones are in attendance. If Smith attends, then Jones will too. Jones will not be in attendance. Therefore, the deal will not be completed. **Solution**: First, we rephrase the argument in premise conclusion form: 1. The deal will be completed only if Smith or Jones are in attendance. 2. If Smith attends, then Jones will too. 3. Jones will not be in attendance. 4. The deal will not be completed. Second, we translate the argument into the language of propositional logic: 1. $p \to (q_1 \vee q_2)$2. $q_1 \to q_2$ 3. $\neg q_2$4. $\neg p$. **Translation key**: p: The deal will be completed. q_1: Smith is in attendance. q_2: Jones is in attendance. Third, we use the search for counterexample method to check whether there is an assignment on which the premises come out true and the conclusion false:

p	q_1	q_2	$p \to (q_1 \vee q_2)$	$q_1 \to q_2$	$\neg q_2$	$\neg p$
T	F	F	!	T	T	F

Since there is no assignment on which the premises come out true and the conclusion false, we conclude that the argument is propositionally valid.

b. All logicians are philosophers. All logicians learn propositional logic at some point. So, some philosophers learn propositional logic at some point. **Solution**: First, we rephrase the argument in premise conclusion form: 1. All logicians are philosophers. 2. All logicians learn propositional logic at some point. 3. Some philosophers learn propositional logic at some point. Second, we translate the argument into the language of propositional logic: 1. p2. q 3. r **Translation key**: p: All logicians are philosophers. q: All logicians learn propositional logic at some point. r: Some philosophers learn propositional logic at some point. Third, we note that there is a simple an assignment on which the premises come out true and the conclusion false:

p	q	r	p	q	r
T	T	F	T	T	F

c. The deal will not be completed unless someone is in attendance. Neither Smith nor Jones will be in attendance. Therefore, the deall will not be completed.

3. Identify occurrences of designators, predicates, and quantifier expressions in the following sentences:

a. USC is located in Los Angeles. **Solution**: $\underbrace{\text{USC}}_{\text{designator}}$ $\underbrace{\text{is located in}}_{\text{predicate}}$ $\underbrace{\text{Los Angeles}}_{\text{designator}}$

b. Some private university is located in NYC.

c. Some critic admires every critic. **Solution**: $\underbrace{\text{Some critic}}_{\text{quantifier phrase}}$ $\underbrace{\text{admires}}_{\text{predicate}}$ $\underbrace{\text{every critic}}_{\text{quantifier phrase}}$

d. Mark Twain wrote *Tom Sawyer*.

e. Every students owns a computer. **Solution**: $\underbrace{\text{Every student}}_{\text{quantifier phrase}}$ $\underbrace{\text{owns}}_{\text{predicate}}$ $\underbrace{\text{a computer}}_{\text{quantifier phrase}}$

f. Everyone admires someone.

g. Some workday comes after a weekend day. **Solution**: $\underbrace{\text{Some workday}}_{\text{quantifier phrase}}$ $\underbrace{\text{comes after}}_{\text{predicate}}$ $\underbrace{\text{a weekend day}}_{\text{quantifier phrase}}$

6. Quantificational Logic

1. Determine whether each of the following expressions is an atomic formula of quantificational logic.

   a. $\neg R^2(x, y)$

   b. $R^2(x, y)$

   c. $A^3(a, b, x)$ **Solution**: This *is* an atomic formula of quantificational logic. A^3 is a three-place predicate and while a and b are constants, x is a variable.

   d. $Q^1(e)$

   e. $P^4(x, y, z, a)$

   f. $R^2(P^1, a)$ **Solution**: This is *not* an atomic formula of quantificational logic. R^2 is a two-place predicate but P^1 is neither a constant nor a variable.

2. Determine whether each of the following expressions is a formula of quantificational logic.

   a. $\neg\neg R^2(x, y)$

   b. $(P^1(a) \to (A^2(x, y) \wedge \exists x Q^1(x))$ **Solution**: This *is* a formula of quantificational logic. By rule 2, $(P^1(a) \to (A^2(x, y) \wedge \exists x Q^1(x))$ is a formula if $P^1(a)$ and $(A^2(x, y) \wedge \exists x Q^1(x))$ are each a formula. By rule 1, $P^1(a)$ is a formula because it is an atomic formula. By rule 2, $(P^1(a) \to (A^2(x, y) \wedge \exists x Q^1(x))$ is a formula if both $A^2(x, y)$ and $\exists x Q^1(x)$ are formulas. By rule 1, $A^2(x, y)$ is a formula because it is an atomic formula. By rule 3, $\exists x Q^1(x)$ is a formula if $Q^1(x)$ is a formula. By rule 1, both $Q^1(x)$ is a formula because it is an atomic formula. So, we conclude that $(P^1(a) \to (A^2(x, y) \wedge \exists x Q^1(x))$ is a formula.

   c. $\forall x R^2(y, y)$

   d. $\exists xy(P^1 \wedge P^1(y))$

   e. $\forall\forall x R^2(y, x)$ **Solution**: This is *not* a formula of quantificational logic. It is neither an atomic formula nor a negation, conjunction, disjunction, conditional or biconditional. It is finally *not* of the form $\forall v\varphi$ as the first quantifier is not followed by a variable. By rule 4, we conclude that it is not a formula of quantificational logic.

   f. $\exists x \exists y R(x, y)$

3. Determine whether each of the following expressions is an *abbreviation* for a formula of quantificational logic.

a. $\exists x \exists y R(x, y) \rightarrow P(x)$ **Solution**: This is an abbreviation for the formula $(\exists x \exists y R^2(x, y) \rightarrow P^1(x))$.

b. $\forall x R(x) \vee \exists y Q(y, y)$

c. $P^1(x) \wedge Q^1(x) \vee R^2(x, y)$ **Solution**: This is *not* an abbreviation for a formula of quantificational logic.

d. $P^1(x) \wedge Q^1(x) \rightarrow R^2(x, y)$

e. $\exists x P(x) \wedge \exists y Q(y) \wedge \exists z R(z)$

f. $\exists x \forall y \exists z (P(x) \wedge Q(y) \wedge R(x, y, z))$ **Solution**: This is an abbreviation for the formula $\exists x \forall y \exists z ((P^1(x) \wedge Q^1(y) \wedge R^3(x, y, z))$.

4. Determine whether the following formulas contain a free occurrence of the variable x.

a. $\forall x (R(x, y) \wedge R(y, x))$

b. $\forall x R(x, y) \wedge R(y, x)$ **Solution**: By rule 1, the occurrence of x in $R(y, x)$ is free. By rule 2, the occurrence of x in $R(y, x)$ remains free in the conjunction $\forall x R(x, y) \wedge R(y, x)$.

c. $(\exists x (P(x) \wedge Q(x)) \rightarrow \forall y R(x, y))$ **Solution**: By rule 1, the occurrence of x in $R(x, y)$ is free. By rule 2, the occurrence of x in $\forall y R(x, y)$ remains free in the conditional $\exists x (P(x) \wedge Q(x) \rightarrow \forall y R(x, y)$.

d. $\forall x \exists y R(x, y) \rightarrow \exists x \forall y R(y, x)$

e. $(\forall x \exists y R(x, y) \rightarrow \forall y R(y, x))$

f. $\forall x (\exists y R(x, y) \rightarrow \forall y R(y, x))$

5. Determine whether the following formulas are *open* or *closed*. Justify your answers.

a. $\forall x R(x, y) \wedge R(y, x, z)$ **Solution**: This is an *open* formula as the occurrences of the variables y and z as well as the last occurrence of the variable x are all free.

b. $R(a, b)$ **Solution**: This is a *closed* formula, since no variables occur, much less freely, in it.

c. $(\exists x (P(a, x) \wedge Q(x, a)) \rightarrow \forall y R(x, y))$

d. $\forall x R(y, y)$ **Solution**: This is an *open* formula as the two occurrences of the variables y both free.

e. $\forall x \exists y \forall z R(x, y, x)$

f. $(R(a, x) \rightarrow \forall x R(x, x))$

6. Determine which of the following formulas are true in the following model:

- $D = \{p_1, p_2, p_3, p_4\}$
- $a : p_1$
- $b : p_2$
- P_1 : enclosed by a circle
- Q : enclosed by a dark grey figure
- P_2 : enclosed by a diamond
- R : points to

a. $R(a, b)$ **Solution**: This *not* true in the model because it is true relative to all assignments. Whatever the assignment, the denotation of a in the model, i.e., p_1, does *not* point to the denotation of b in the model, i.e., p_2.

b. $R(a, b) \rightarrow R(b, b)$

c. $P(a) \wedge Q(b)$

d. $\exists x P_1(x) \wedge \exists x P_2(x)$

e. $\forall x(\exists y R(x, y) \vee \exists y R(y, x))$ **Solution**: This formula is true in the model because it is true relative to all assignments. Given an assignment α, we verify that for every $p \in D$, the formula $\exists y R(x, y) \vee \exists y R(y, x)$ is true relative to $\alpha[x/p]$. There are four cases to consider: $\alpha[x/p_1]$: $\exists y R(x, y)$ is true in the model relative to this assignment because there is a member of the domain, p_3, such that $R(x, y)$ is true relative to the assignment $\alpha[x/p_1][y/p_3]$. This is because the denotation of x relative to this assignment, p_1, points to the denotation of y under the assignment, p_3. Therefore, the disjunction $\exists y R(x, y) \vee \exists y R(y, x)$ is true relative to $\alpha[x/p_1]$. $\alpha[x/p_2]$: $\exists y R(x, y)$ is true in the model relative to this assignment because there is a member of the domain, p_4, such that $R(x, y)$ is true relative to the assignment $\alpha[x/p_2][y/p_4]$. This is because the denotation of x relative to this assignment, p_2, points to the denotation of y under the assignment, p_4. Therefore, the disjunction $\exists y R(x, y) \vee \exists y R(y, x)$ is true relative to $\alpha[x/p_2]$. $\alpha[x/p_3]$: $\exists y R(y, x)$ is true in the model relative to this assignment because there is a member of the domain, p_1, such that $R(y, x)$ is true relative to the assignment $\alpha[x/p_3][y/p_1]$. This is because the denotation of y relative to this assignment, p_1, points to the denotation of x under the assignment, p_3. Therefore, the disjunction $\exists y R(x, y) \vee \exists y R(y, x)$ is true relative to $\alpha[x/p_3]$. $\alpha[x/p_4]$: $\exists y R(y, x)$ is true in the model relative to this assignment because there is a member of the domain, p_2, such that $R(y, x)$ is true relative to the assignment $\alpha[x/p_4][y/p_2]$. This is because the denotation of y relative to this assignment, p_2, points to the denotation of x under the assignment, p_4. Therefore, the disjunction $\exists y R(x, y) \vee \exists y R(y, x)$ is true relative to $\alpha[x/p_4]$.

f. $\forall x(P_1(x) \to \exists y(Q(y) \wedge R(x, y)))$

7. Translation into Quantificational Logic

1. Translate the following sentences into the language of quantificational logic.

a. Some Trojans fear Achilles, but Achilles fears every Trojan. **Solution**:

$$\exists x(P(x) \wedge R(x, a)) \wedge \forall x(P(x) \to R(a, x))$$

Translation key: a: Achiles P: Trojan R: fears

b. All Trojans are courageous. **Solution**:

$$\forall x(P(x) \to Q(x))$$

Translation key: P: Trojan Q: courageous.

c. Only Trojans come from Troy. **Solution**:

$$\forall x(R(x, a) \to P(x))$$

Translation key: a: Troy P: Trojan R: comes from

d. No Greek comes from Troy.

e. Achilles fears Paris, but Paris fears no one. **Solution**:

$$R(a, b) \wedge \forall x \neg R(b, x)$$

Translation key: a: Achiles b: Paris R: fears

f. Some Greeks fears some Trojans, and some Trojans fear some Greeks.

2. Assess the following arguments for quantificational validity.

a. All poisonous frogs are amphibians. Kermit is a poisonous amphibian. Therefore, Kermit is a frog. **Solution**: The argument is quantificationally *invalid*.1. $\forall x((P(x) \wedge Q(x)) \to R(x))$2. $P(a) \wedge R(a)$3. $Q(a)$ **Translation key**: a: Kermit P: poisonous Q: frog R: amphibianThe following model makes the premises true and the conclusion false. D: $\{p_1, p_2\}$$a$: p_1 P: enclosed by a circle Q: enclosed by a diamond R: grey

b. Aristotle admired anyone who taught for free. Socrates taught for free, but no sophist did ever teach for free. So, Aristotle did not admire a sophist.

c. No weekday comes after Saturday. Monday is a weekday, but Sunday is not. Therefore, Monday does not come after Sunday. **Solution**: The argument is quantificationally *invalid*.1. $\forall x(P(x) \to \neg R(x,a))$2. $P(b) \land \neg P(c)$3. $\neg R(c,b)$ **Translation key**: a: Saturday b: Monday c: Sunday P: weekday R: comes afterThe following model makes the premises true and the conclusion false. D: $\{p_1, p_2, p_3\}a$: p_1 P: enclosed by a circle R: points to

d. Some cube is green all over and some cube is red all over. No cube is both green all over and red all over. Some sphere is green all over. So, some sphere is not red all over.

e. No Trojan fears Achilles, but Achilles fears some Trojans. Paris is a Trojan. So, Achilles fears Paris but Paris does not fear Achilles. **Solution**: The argument is quantificationally *invalid*.1. $\forall x(P(x) \to \neg R(x,a)) \land \exists x(P(x) \land R(a,x))$2. $P(b)$3. $R(a,b) \land \neg R(b,a)$ **Translation key**: a: Achilles b: Paris P^1: Trojan R^2: fearsThe following model makes the premises true and the conclusion false. D: $\{p_1, p_2\}a$: p_1b: p_2P: enclosed by a circle$R$: points to

f. Some stadium is close to University Park. The Coliseum is a stadium. Therefore, the Coliseum is close to University Park.

g. Los Angeles is connected by direct flight to London. London is connected by direct flight to every major European city. Barcelona is a major European city. Therefore, Los Angeles is connected by direct flight to Barcelona. **Solution**: The argument is quantificationally *invalid*.1. $R(a,b)$2. $\forall x(P(x) \to R(b,x))$3. $P(c)$4. $R(a,c)$ **Translation key**: a: Los Angeles b: London c: Barcelona P: major European city R: connected by direct flightThe following model makes the premises true and the conclusion false. D: $\{p_1, p_2, p_3\}a$: p_1b: p_2c: p_3P: enclosed by a circle$R$: points to

h. All Greeks feared Hector or Paris. Paris feared no Greek who feared Hector. Therefore, some Greeks feared Paris.

8. Natural Deduction for Quantificational Logic

1. Provide a natural deduction proof in order to justify each of the claims below:

   a. $\exists x P(x) \to \exists x Q(x), \forall x P(x) \vdash \exists x Q(x)$

b. $\exists x P(x) \wedge \exists x Q(x), \neg\exists x(P(x) \wedge Q(x)), \forall x(R(x) \to P(x)) \vdash \exists x \neg R(x)$
Solution:

c. $\forall x(P(x) \to \neg Q(x)), \forall x(R(x) \to Q(x)) \vdash \forall x(R(x) \to \neg P(x))$

d. $\forall x(P(x) \to (\exists y R(x,y) \to Q(x))), \neg\exists y Q(y) \vdash \forall y(P(y) \to \forall z \neg R(y,z))$ **Solution**:

e. $\forall x(P(x) \to \forall y(R(x,y) \to P(y))), \exists x(Q(x) \wedge P(x)) \vdash \exists x(Q(x) \wedge \forall y(R(x,y) \to P(y)))$ **Solution**:

10. Identity

1. For each set of formulas below, find a model we can use to justify the fact that the set is consistent. Please justify your answers.

   a. $\{\forall x(P(x) \leftrightarrow Q(x)), P(a), Q(b), \neg a = b\}$

   b. $\{\forall x(R(a,x) \to R(b,x), \neg R(a,a), R(b,a)\}$

2. For each argument below, find a model we can use to justify the fact that the argument is invalid. Please justify your answers.

   a. $\forall x(P(x) \to Q(x)), \neg Q(a) \vdash a = b \to R(a,b)$

   b. $\exists x P(x), \exists x Q(x) \vdash \exists x \exists y \neg x = y$

3. Provide a natural deduction proof to justify each of the claims below:

   a. $\forall x(P(x) \to Q(x)), \neg Q(a), P(b) \vdash a = b \to R(a,b)$

   b. $\exists x \forall y(P(x) \leftrightarrow y = x), P(a) \vdash \forall x(P(x) \to x = a)$

   c. $\exists x P(x), \exists x Q(x), \neg\exists x(P(x) \wedge Q(x)) \vdash \exists x \exists y \neg x = y$